

Knowledge Creation and Diffusion with Limited Appropriation*

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Abstract

Innovation is central to economic growth, but so is the diffusion of new knowledge. Such is the finding of recent macro papers that model the interaction between these two forces. Absent in this literature are three key elements that are the focus of this paper. First, we consider the role of frictions in matching innovators and imitators, which mediate the process of knowledge transmission. Second, we introduce the force of creative destruction, whereby innovators are replaced by imitators. Third, while most of the recent literature has focused on the case where all surplus from knowledge transmission is captured by the imitators, we consider the full range of surplus shares that the innovators and imitators can appropriate and their impact on growth. In a simple one-period model, we derive a modified Hosios condition for the optimal appropriation when firms are homogeneous. But we also find that as the degree of heterogeneity increases, the share of innovators must decrease to maximize growth. Our calibrated dynamic model suggests that the optimal share of surplus that innovators appropriate should lie in the medium range.

Keywords: Innovation, knowledge diffusion, endogenous growth, appropriation

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1 Introduction

Knowledge creation and diffusion are the main forces behind economic growth. Starting with [Lucas \(2009\)](#), a series of recent papers in the macro literature have emphasized the role of diffusion as a contagion process where lagging firms learn the knowledge of more advanced firms in random matching. [Moll and Lucas \(2014\)](#) and [Perla and Tonetti \(2014\)](#) are examples. In these papers the surplus generated is assumed to accrue entirely to the firm on the receiving end. In contrast, many forms of knowledge transmission involve some degree of appropriation by the transferring firms, such as technology licensing deals. Our paper captures this feature by modeling knowledge transfer as a bargaining problem between these two parties. More explicitly, we consider the role of the Nash bargaining weights in this problem and their impact on knowledge creation and transmission.

Intellectual property rights are a critical part of innovation policy. There is a long literature arguing in favor or against strong patent rights, trading off incentives for innovation and its costs. In our setting, stronger patent rights are associated with a higher bargaining weight for innovators. Our paper considers the impact this has on both incentives for innovation and learning, and on its overall effect on economic growth. We explore this both theoretically and quantitatively, calibrating a model of innovation and knowledge diffusion.

An increase in the bargaining weight of the innovator, and thus its share of the surplus, has two opposing effects: 1) it directly encourages innovation; and 2) it discourages learning. In the presence of congestion a la [Mortensen and Pissarides \(1994\)](#), this in turn reduces the contact rate for innovators (increases for imitators), having a negative effect on innovation. At one extreme, when all surplus is appropriated by the firm transferring knowledge, thus holding up the learning firm, there are no incentives for learning and knowledge transfer disappears. At the other extreme, when the learning firm appropriates all surplus, innovation occurs only for its direct productive benefit to the innovator who disregards the value created by knowledge transfer.

Intellectual property rights play a dual role. On the one hand, they allow innovators to appropriate some of the rents from follow-up firms that use or build upon these innovations. On the other hand, they can discourage competing innovations that follow up and destroy some of the rents obtained by the original innovator, i.e. creative destruction. The relative importance of each of these channels depends on the share of creative destruction as a fraction of total innovation. We find that this plays an important role when considering the impact of intellectual property rights on economic growth.

In a simple one-period model where all firms are ex-ante identical, we show that the

maximum level of growth is achieved at an intermediate bargaining weight for innovators that is the one suggested by the well-known Hosios condition ([Hosios, 1990](#)) in models of random matching.¹ When firms differ in their initial level of productivity, those above a certain threshold choose to innovate while those below choose to learn from the innovators. We find that as ex-ante heterogeneity increases, the optimal bargaining weight for innovators decreases and becomes zero when heterogeneity is sufficiently high. We also find that the optimal bargaining weight increases with the share of creative destruction, measured by the extent to which adoption by a learning firm decreases the original innovator's value.

Heterogeneity matters for two reasons. Firstly, it corresponds to a component of knowledge that is exogenous to innovation efforts of a firm. When the importance of this component overwhelms the one resulting from innovation effort, stronger IP modeled here as higher innovator share of surplus gives rents to innovators without significantly affecting their innovation or participation decisions, while discouraging imitation and learning at the same time. Secondly, with ex-ante heterogeneity the marginal type (the one at the threshold) represents a less valuable match for imitators to learn from. Thus, the positive external effect on imitators is smaller, while the negative crowding out effect on innovators is the same.

We explore the question quantitatively in a dynamic general equilibrium scenario that is a small variation of the basic setup in [Benhabib, Perla and Tonetti \(2021\)](#) with the addition of matching congestion. There is a fixed set of agents that can either operate as productive firms or learn from others. Each firm's productivity follows a Brownian motion where drift is a function of costly innovation effort and volatility is given. Learning agents contact a randomly selected firm at a rate that is determined via a matching function. Upon meeting, the two agents split the value of the learning firm according to Nash bargaining. With some probability, the original innovator loses the value of its innovation and goes back to join the pool of learning agents. This parameter measures the relative share of creative destruction.

The model is calibrated to match a series of aggregate moments.² Given parameter

¹The intuition for this result is as follows. Holding fixed the innovation decision μ , the Hosios condition guarantees that the equilibrium delivers the optimal fraction of innovators in the population, i.e. the one that maximizes total surplus. What is more surprising, is that total surplus when also taking into account the equilibrium choice of μ is also maximized at this point. The intuition behind this result is that in equilibrium innovator profits from innovation are proportional to total surplus, when surplus is maximized so are the total (direct and indirect) incentives for innovation.

²These moments include: aggregate growth rate, interest rate, size distribution of firms, volatility of firm size, share of creative destruction (as given by [Garcia-Macia, Hsieh and Klenow \(2019\)](#)) and the ratio of public to private returns to innovation (as given by [Bloom, Schankerman and Van Reenen \(2013\)](#)).

values, we calculate the share of surplus that maximizes growth. As in our simple model, we find that as volatility becomes large and heterogeneity increases, the optimal innovator share of surplus decreases. In our preliminary calibration, where volatility is chosen to match moments of the size distribution of firms, the optimal share is in the medium range. Given these parameter values, total growth falls with innovator share mostly because of the severe drop in knowledge transmission, as the fraction of learning firms decreases dramatically with the bargaining weight of innovators. The optimal fraction of innovators is also sensitive to parameters in the matching function. As we vary the elasticity of the matching function from zero to one, the optimal fraction of innovators also goes from zero to one. This shows the importance of matching frictions in the process of knowledge transfer when considering the optimal assignment of intellectual property rights.

1.1 Related Literature

Our paper builds on several strands of literature, including the papers on knowledge diffusion, patent policy and innovation, and matching frictions, as well as recent papers attempting to measure returns to innovation and the extent of creative destruction.

[Kortum \(1997\)](#) considers a setting where firms sample from a fixed distribution of ideas, recognizing that a key to sustained exponential growth is that the stationary part of this distribution has a Pareto upper tail. In [Luttmer \(2007\)](#), this distribution is endogenously generated in the steady state, as entrants learn from the distribution of incumbent’s productivity. A similar mechanism is developed in [Lucas \(2009\)](#), where all firms exogenously learn from others and the initial distribution of knowledge has a Pareto tail.³ [Moll and Lucas \(2014\)](#) and [Perla and Tonetti \(2014\)](#) endogenize the allocation of resources devoted to learning. In these models, resources can be allocated either to production or learning.

Our model builds on [König, Lorenz and Zilibotti \(2016\)](#) and [Benhabib et al. \(2021\)](#), where firms make an optimal choice between innovation and imitation. In a contemporaneous extension of their paper, [Benhabib et al. \(2021\)](#) briefly consider the role of bargaining. Our analysis differs in several respects: First, we focus on the question of intellectual property protection and derive explicitly the bargaining weights from the strength of patent enforcement. Second, we focus on the relationship between patent enforcement and growth, and how the growth maximizing policy is affected by several characteristics of the environment, including innovator heterogeneity, matching frictions, and creative destruction. The latter two characteristics are new features we introduce and

³A class of hybrid models where firm productivity can evolve from the sampling of an exogenous distribution (as in [Kortum \(1997\)](#)) and the distribution of other firms’ knowledge (as in [Luttmer \(2007\)](#) and [Lucas \(2009\)](#)) can be found in [Alvarez, Buera and Lucas \(2008\)](#) and [Buera and Oberfield \(2020\)](#).

absent in their model.

We borrow from recent papers that quantify the technology and product market spillover effects of innovation. Mapping our model to their framework, we use [Bloom et al. \(2013\)](#)'s calculation of the social versus private returns to innovation. Similarly, we use the importance of creative destruction measured by [Garcia-Macia et al. \(2019\)](#).

Our insights on the optimal bargaining weights relate to the literature on matching frictions in random search. Using an alternative protocol relative to Nash bargaining, our expression for the optimal weights resembles [Hosios \(1990\)](#)'s condition, by adjusting for the creative destruction events. In the presence of heterogeneity, however, this condition fails as pointed out by [Shimer and Smith \(2001\)](#).

2 An Illustrative Model

We first deploy a simple one-period model to illustrate the key trade-offs in the incentives for innovation and imitation. The primitives of the environment described here carry through to our dynamic model.

2.1 A Search Market for Knowledge Transfers

The economy lasts for one period and consists of a unit mass of firms. Firms are endowed with an initial knowledge denoted by z , which follows a cumulative distribution function $F(z)$. They can engage in either of two activities, which we call innovation or imitation, that result in an end-of-period knowledge z' . Firms obtain a productive value equal to their final knowledge.

A two-stage game takes place, an innovation stage followed by an imitation stage. In the innovation stage, firms choose whether and how much to innovate. They can improve their productivity to $z' = \mu z$ by incurring an investment cost $c(\mu)z$. The innovation cost function is convex and strictly increasing in the innovation intensity μ . The proportion of firms that invest in innovation is denoted by α , and the remaining $1 - \alpha$ of firms wait to imitate.

In the imitation stage, innovators and imitators match randomly in the market for knowledge transfer. The number of matches is given by the matching function $M(\alpha, 1 - \alpha)$, which is increasing in both arguments and has constant returns to scale. Imitators meet a firm randomly selected from the pool of innovators, with a meeting probability $\iota(\alpha) \equiv M/(1 - \alpha)$. Consistently, innovators meet an imitator with probability $\lambda(\alpha) \equiv M/\alpha$. If an imitator meets an innovator of knowledge z' , it can copy the innovator's knowledge

perfectly; otherwise, its productive value is zero.⁴ The technology transfer thus generates a surplus of z' . The innovator bargains for a license fee with a bargaining weight β . Thus, the innovator appropriates a share β of the surplus, leaving a share $1 - \beta$ to the learning firm.

Congestion in Knowledge Diffusion. The matching process in the knowledge transfer market features potential congestion in learning. If there are more imitators searching for better technologies to copy, it becomes more difficult for them to find such opportunities. This could represent the limited ability of innovators to transfer their knowledge. To illustrate why congestion matters, consider a special case in which the probability of meeting an innovator is constant, and hence independent of the share of innovators, as in the knowledge diffusion models by [Perla and Tonetti \(2014\)](#), [Buera and Oberfield \(2020\)](#), [Perla, Tonetti and Waugh \(2021\)](#), and [Benhabib et al. \(2021\)](#). This situation corresponds to a matching function $M(\alpha, 1 - \alpha) = \kappa(1 - \alpha)$, where an infinitesimally small mass of innovators can spread their knowledge to the whole economy.

Bargaining Weights and Patent Enforcement. The innovator's bargaining weight represents the patent enforcement probability under the intellectual property rights regime. Stronger patent protection in favor of innovators, indicated by higher enforcement, grants innovators greater bargaining power and allows them to appropriate a larger share of the rents generated from knowledge diffusion.

We consider the following *enforcement game*. Upon being matched, the innovator makes a take-it-or-leave-it offer to the imitator, asking for a license fee t for the technology transfer. If the imitator rejects the offer and proceeds to copy the technology without obtaining a license, it gets caught with probability β . Once caught, the imitator is excluded from using this technology and receives its outside value of zero. Consequently, the innovator faces a participation constraint for the imitator $z' - t \geq (1 - \beta)z'$. This constraint binds when the innovator asks for a license fee $t = \beta z'$.

Equilibrium. Innovators choose innovation intensity μ to maximize their payoff:

$$\max_{\mu} \left[(1 + \lambda(\alpha)\beta)\mu - c(\mu) \right] z.$$

The payoff of innovation comes in two parts: a direct payoff through increased output, μ , and an indirect payoff through knowledge transfer, $\beta\lambda(\alpha)\mu$. Their optimal innovation

⁴In the dynamic model, we generalize the learning technology allowing for imperfect copying.

decision equates the marginal private payoff to the marginal cost:

$$1 + \lambda(\alpha)\beta = c'(\mu). \quad (1)$$

All else being equal, a higher chance of meeting imitators or a higher ability to appropriate rents from knowledge transfers will spur more innovation.

More productive firms find it more appealing to innovate on their own, while less productive firms have more to gain from imitation and, therefore, choose to do so. The marginal firm who is indifferent between innovating or imitating is denoted by a threshold productivity $\underline{z}$, where the resulting fraction of innovating firms $\alpha = 1 - F(\underline{z})$. This marginal firm is characterized by an indifference condition:

$$\left[(1 + \lambda(\alpha)\beta) \mu - c(\mu) \right] \underline{z} = \iota(\alpha) (1 - \beta) \mu \mathbb{E} [z | z \geq \underline{z}]. \quad (2)$$

Given the enforcement regime β , an equilibrium consists of an allocation $\{\alpha, \mu\}$ that satisfies equations (1) and (2).⁵

2.2 Optimal Appropriation

There are several sources of externality that may result in suboptimal equilibrium outcomes. One source is the incomplete appropriation of rents from knowledge transfers, which can lead to firms exerting too little investment. Another source is congestion externality in the matching process, closely related to those found in labor search models. Both are affected by the strength of patent enforcement. A stronger enforcement β improves the incentives to innovate, holding fixed the composition of innovators and imitators in the population. At the same time, it discourages imitation, leading to a higher fraction of innovators and lowering the frequency of knowledge transfers for innovators.

With full enforcement $\beta = 1$, innovators appropriate all surplus. As a result, the incentives for learning and knowledge transfer disappear, decreasing the returns to innovation. Conversely, with no enforcement $\beta = 0$, the learning firm appropriates all surplus, so innovators gain nothing from knowledge transfers. Innovation is minimized at both extremes. It follows immediately that the first-best level of innovation cannot be achieved with one policy instrument.

To correct for these externalities, the first-best allocation can potentially be achieved with a sufficiently rich set of instruments, for example with the appropriate strength

⁵The equilibrium is unique and characterized by an interior allocation, i.e., the fraction of innovators $\alpha \in (0, 1)$.

of patent protection combined with direct subsidies to innovation. However, as often discussed in the patent literature, direct subsidies might be difficult to implement when innovation is not directly observed (see [Kremer, 1998](#)). For these reasons, we ask what can be achieved when the planner has at its disposal only the strength of patent enforcement. This policy instrument β is chosen to maximize social welfare subject to the equilibrium incentives for innovation and imitation.

Formally, the planner chooses β and, correspondingly, the constrained optimal allocation, $\{\alpha, \mu\}$, that maximizes total output net of innovation costs:

$$\max_{\beta, \alpha, \mu} \underbrace{\alpha \mathbb{E} [z | z \geq \underline{z}] (\mu - c(\mu))}_{\text{output from incumbents}} + \underbrace{M(\alpha, 1 - \alpha) \mathbb{E} [z | z \geq \underline{z}] \mu}_{\text{surplus from matching}} \quad (3)$$

subject to the equilibrium conditions (1) and (2).

In the calculation of (3), the social surplus from knowledge diffusion per match depends not only on the extent of investment μ but also on the selection into innovation—the average innovator knowledge $\mathbb{E} [z | z \geq \underline{z}]$.

Equation (2) implies that α is a strictly increasing function of β ; this defines implicitly β and μ as functions of α .⁶ Maximizing the planner's objective (3) with respect to α gives:

Proposition 1. *The optimal innovator appropriation is characterized by:*

$$\underbrace{(\omega - \beta) \left(\underline{z} + \frac{\alpha}{1 - \alpha} \mathbb{E} [z | z \geq \underline{z}] \right)}_{\text{Congestion}} - \underbrace{(1 - \omega) (\mathbb{E} [z | z \geq \underline{z}] - \underline{z})}_{\text{Displacement}} + \underbrace{(1 - \beta) \mathbb{E} [z | z \geq \underline{z}] \mathcal{E}_{\mu, \alpha}}_{\text{Innovation Spillover}} = 0, \quad (4)$$

where the matching elasticity $\omega \equiv \frac{\partial \log(M)}{\partial \log(\alpha)}$ and the innovation elasticity $\mathcal{E}_{\mu, \alpha} \equiv \frac{\partial \log(\mu)}{\partial \log(\alpha)}$.

Congestion. The first component in equation (4) captures the congestion externality in the matching market. Placing one additional firm in the pool of innovators reduces the matching probability for the existing innovators, as captured by the matching elasticity ω . This congestion externality is properly internalized when we assign a bargaining weight $\beta = \omega$ to the innovators, as prescribed by the standard [Hosios \(1990\)](#)'s condition.

Introducing matching congestion in knowledge diffusion matters and can be crucial for the result. Again it is instructive to consider an economy with no congestion on the imitator's side, and the chance of finding an innovator to copy is independent of the composition of innovators and imitators. This corresponds to maximum congestion

⁶This function is defined, increasing and differentiable from α_0 to $\alpha = 1$ where α_0 is the share of innovators when $\beta = 0$.

in the innovator side, i.e., $\omega = 0$. The optimal innovator appropriation β would then approach zero. From equation (2) it also follows that the innovator's meeting rate $\lambda(\alpha)$ must diverge to ∞ , so only an infinitesimally small fraction of innovators are needed to spread the knowledge to the whole economy.

Displacement. The entry of a marginal type $\underline{z}$ lowers the value of knowledge diffusion per match, as it displaces better learning opportunities whose expected value is $\mathbb{E}[z|z \geq \underline{z}]$. The second term in equation (4) captures this force and indicates that weakening the innovator's bargaining power can deter low-type entrants, thereby improving the overall knowledge pool. The strength of the displacement externality depends on the degree of heterogeneity, which we measure by either the gap $\mathbb{E}[z|z \geq \underline{z}] - \underline{z}$ or the mean-to-margin ratio $\mathbb{E}[z|z \geq \underline{z}] / \underline{z}$.

This displacement effect is also shaped by the degree of congestion. When matching is uncongested on the innovator side ($\omega = 1$), the entry of a marginal type does not crowd out the average existing matches, and the displacement effect vanishes. As congestion rises ($\omega < 1$), however, the arrival of a marginal entrant increasingly replaces higher-quality learning opportunities, amplifying the negative impact of low-type entry on the average quality of knowledge diffusion.

The congestion externality alone suggests that innovators should be compensated according to their share of the contribution to match creation. The displacement externality indicates that the surplus division should be adjusted to reflect their contribution to surplus creation. This modification aligns with the generalized Hosios condition characterized by [Mangin and Julien \(2021\)](#), which they note applies across a wide range of search models.

Innovation Spillover. The bargaining weight β affects the incentives for innovation in two ways. One is the direct effect, which encourages firms to invest. The second is an indirect effect, since higher β will lead to an increase in α and, consequently, lower contact rates for innovators. The net effect is non-monotonic: at the extremes, when β is zero, there are no gains to innovators from knowledge transfer, and when $\beta = 1$, incentives for learning disappear, so diffusion shuts down.

The value of β (or equivalently α) that maximizes the innovation spillover depends on details of the productivity distribution and innovation technology, except when the mean-to-margin ratio $\mathbb{E}[z|z \geq \underline{z}] / \underline{z}$ does not depend on the composition α , as in the case of a Pareto distribution or when there is no heterogeneity. In such a situation, the innovation spillover is maximized when $\beta = \omega$, as in the Hosios condition.

Table 1: Heterogeneity and optimal bargaining weight

ζ	Mean/Margin $\frac{\mathbb{E}[z]}{z}$	Optimal innovator appropriation β			Innovator share α	% Output diffusion
		Congestion	– Displacement	+ Spillover		
20	1.05	0.5	0.5	0.5	82%	35%
2	2	0.5	0.28	0.36	47%	57%
1.1	11	0.5	0	0.13	1%	92%

Notes: The matching function is $M(\alpha, 1 - \alpha) = \kappa \alpha^\omega (1 - \alpha)^{1-\omega}$, with $\kappa = 0.5$ and $\omega = 0.5$. The innovation cost function is $c(\mu) = \frac{\chi}{1+\varepsilon} \mu^{1+\frac{1}{\varepsilon}}$, with $\chi = 2$ and $\varepsilon = 1$. The creative destruction probability is $\delta = 0$.

2.3 Heterogeneity and Displacement: A Pareto Distribution Example

To illustrate the impact of heterogeneity on the optimal bargaining weight, we construct a simple numerical example. The productivity level follows a Pareto distribution: $\forall z \in [\underline{z}, \infty)$, $F(z) = 1 - \left(\frac{\underline{z}}{z}\right)^\zeta$, where the tail index $\zeta > 1$. A lower ζ implies a fatter tail in the productivity distribution and a greater extent of heterogeneity. As mentioned earlier, regardless of the number of firms participating in innovation, the average quality of innovator pool compared to the marginal one remains constant at $\zeta/(\zeta - 1)$.

Table 1 presents the relationship between optimal innovator bargaining power and heterogeneity within the economy. The matching function employed in our analysis exhibits a constant elasticity, denoted as $\omega = 0.5$. Our findings indicate that higher levels of heterogeneity are associated with reduced optimal innovator bargaining power. Specifically, when there is minimal heterogeneity, as observed in the scenario where $\zeta = 20$ and the Pareto tail is thin, the optimal bargaining weight aligns closely with the standard Hosios level. However, as heterogeneity intensifies, the displacement externality becomes more pronounced. Furthermore, when gains derived from the diffusion of existing knowledge are substantial, the optimal bargaining weight experiences a significant downward adjustment. This effect is evident in the specific example where ζ approaches a value close to 1, ultimately driving the optimal bargaining weight down to zero.

Table 2 further illustrates how the displacement effect varies with the degree of congestion. As congestion intensifies, the optimal bargaining weight for innovators increasingly diverges from the value implied by congestion alone. For instance, when $\omega = 0.7$, the optimal weight is approximately 0.65, very close to 0.7; when $\omega = 0.3$, the optimal weight falls to 0.05, substantially below 0.3. This widening gap reflects the strengthening of the displacement effect under more congested matching, as the entry of additional low-type innovators exacerbates quality losses in the pool of learning opportunities.

These numerical results will also help us to interpret our subsequent results in the

Table 2: Displacement effect and congestion

Congestion ω	Optimal appropriation β	Innovator share α	% Output diffusion
0.3	0.05	13%	80%
0.5	0.36	47%	57%
0.7	0.65	70%	51%

Notes: The matching function is $M(\alpha, 1 - \alpha) = \kappa \alpha^\omega (1 - \alpha)^{1-\omega}$, with $\kappa = 0.5$. The firms follow a Pareto distribution with a tail parameter $\zeta = 2$. The innovation cost function is $c(\mu) = \frac{\chi}{1+\frac{1}{\varepsilon}} \mu^{1+\frac{1}{\varepsilon}}$, with $\chi = 2$ and $\varepsilon = 1$. The creative destruction probability is $\delta = 0$.

dynamic model, where the heterogeneous level of firm productivity is an endogenous outcome from forces of innovation and diffusion.

3 Dynamic Model

We now build a dynamic model of knowledge creation and diffusion. This dynamic model differs from the static one in that the extent of knowledge heterogeneity is an endogenous outcome of firm innovation and learning. Furthermore, it embeds the key elements of a workhorse endogenous innovation model and is, therefore, suitable for quantification.

3.1 Environment

Time is infinite and continuous, $t \in [0, \infty)$. The economy consists of a measure-one continuum of firms, each produces a differentiated intermediate good. The firms are characterized by their productivity z , and $F(z, t)$ is the cumulative distribution of active producing firms. The final consumption good is a CES aggregation of the intermediate goods:

$$Y(t) = \left[\int y(z, t)^{\frac{\eta-1}{\eta}} dF(z, t) \right]^{\frac{\eta}{\eta-1}}, \quad (5)$$

where $y(z, t)$ is the output by a firm with productivity z and η is the elasticity of substitution. Therefore, each producer charges the monopoly markup $\frac{\eta}{\eta-1}$.⁷

There is a representative household with preferences for consumption:

$$\int_0^\infty e^{-\rho t} \log(C(t)) dt.$$

⁷We assume that if there are competitors the next lowest quality competitor is deterred by zero ex post profit under Bertrand competition.

The representative household supplies labor of amount L and owns all the firms in the economy. The implied interest rate in this economy is thus $r(t) = \rho + C'(t)/C(t) = \rho + g(t)$.

Own Innovation. Firms can choose to be in either of two states: active or inactive. If they are active, they can produce and invest to improve their existing productivity through innovation. If inactive, they can search for better technologies and become active when successful. We will call active firms *innovators* and inactive firms, interchangeably, *imitators* or *potential entrants*.⁸

An innovator firm's productivity follows the Brownian motion:

$$dz = \mu dt + \sigma dB, \quad (6)$$

where the drift μ represents the innovation intensity the firm chooses by allocating entrepreneurial time $c(\mu)$ to innovation. Consequently, production uses labor according to

$$y = ((1 - c(\mu)) e^z)^{\frac{1}{\eta-1}} \ell.$$

Knowledge Diffusion. Innovators and imitators match randomly in the market for knowledge transfer as described in Section 2.1. Let $\alpha(t)$ denote the fraction of innovators at time t , where

$$\alpha(t) = 1 - F(\underline{z}(t)), \quad (7)$$

and $\lambda(\alpha(t))$ and $\iota(\alpha(t))$ denote the Poisson contact rates for the innovators and imitators, respectively.

When this meeting occurs, the imitator improves the technology z by a quality ladder $q \in [\underline{q}, \infty)$, which is drawn according to the cumulative distribution function $G(q)$. Thus, the imitator's productivity jumps to $z + q$. With probability $\delta \in [0, 1)$ the knowledge transfer results in “*creative destruction*” of the innovator firm: the imitator learns to produce the same variety; hence, the innovator can be replaced by the imitator, joining the pool of potential entrants. Alternatively, with probability $1 - \delta$, the knowledge transfer leads to a “*new variety*”: the imitating firm can enter with a new variety.

In the event that the imitator learns to produce the same existing variety, the imitator's quality draw needs to be sufficiently good to replace the incumbent, i.e., $q \geq 0$. Thus, the effective creative destruction probability is $\tilde{\delta} = \delta(1 - G(0))$. In contrast, all knowledge

⁸An alternative interpretation is that there exist a unit mass of researchers that can either be investing in improving their technologies or learning from the existing pool. The dichotomy between productive innovating firms and non-productive imitating firms can also be found in the models by [Perla and Tonetti \(2014\)](#) and [Benhabib et al. \(2021\)](#).

transfers that lead to a new variety will take place, so long as the overall quality is above the exit threshold, $z + q \geq \underline{z}(t)$. To summarize, the average quality for entry from creative destruction is higher than for new varieties, as in [Garcia-Macia et al. \(2019\)](#). Here, this outcome arises due to competition and selection.

The innovators and imitators bargain for a license payment for the knowledge transfer according to the enforcement game in Section 2. Therefore, the innovators make a take-it-or-leave-it offer to the imitator, asking for a fee equivalent to β share of the imitator's entry surplus. Appendix A.2 extends the enforcement game to incorporate the learning technology used here and shows that both parties will accept this fee.

3.2 Firm's Problem

We first describe the profits firms obtain from their production activities. A firm producing with technology z derives production profit:

$$\pi(z, t) = \Pi(t) (1 - c(\mu)) e^z,$$

where the aggregate profit level $\Pi(t) = \frac{1}{\eta} Z(t)^{\frac{1}{\eta-1}-1} L$ depends on the aggregate productivity in the economy $Z(t) = \int_{\underline{z}(t)}^{\infty} (1 - c(\mu(z, t))) e^z dF(z, t)$. Further, the final aggregate output is $Y(t) = Z(t)^{\frac{1}{\eta-1}} L$. The details of the firm's static profit maximization and aggregation are relegated to Appendix A.3.

Next, we describe firm decisions in their innovation and imitation activities. Let $V(z, t)$ denote the value of an active innovator firm with productivity z and $W(t)$ the value of an inactive firm at time t . The entrant surplus from copying a firm with knowledge z is

$$S^e(z, t) = \delta \int_{q \geq 0} (V(z + q, t) - W(t)) dG(q) + (1 - \delta) \int_{z+q \geq \underline{z}(t)} (V(z + q, t) - W(t)) dG(q). \quad (8)$$

The innovator value function, $V(z, t)$, and imitator value function, $W(t)$, satisfy the Hamilton-Jacobi-Bellman (HJB) equations:

$$r(t) V(z, t) = \max_{\mu} \left\{ \Pi(t) (1 - c(\mu)) e^z + \mu V_z(z, t) + \frac{1}{2} \sigma^2 V_{zz}(z, t) + V_t(z, t) \right. \quad (9)$$

$$\left. + \lambda(\alpha(t)) \left[\underbrace{\beta S^e(z, t)}_{\text{License Income}} - \underbrace{\tilde{\delta}(V(z, t) - W(t))}_{\text{Creative Destruction Loss}} \right] \right\}$$

$$r(t) W(t) = \iota(\alpha(t)) (1 - \beta) \mathbb{E}[S^e(z, t)] + W_t(t). \quad (10)$$

In equation (9), the flow payoff to the innovating firm consists of two parts: a direct payoff from production profit and an indirect payoff through knowledge transfers. The expected flow of license income is β share of the entry surplus $S^e(z, t)$, taking into account the arrival rate $\lambda(\alpha(t))$. However, with probability $\tilde{\delta}$, knowledge transfers can lead to firms being replaced by new entrants and suffering a loss of $V(z, t) - W(t)$. The first three terms in the first line capture the innovation cost and the resulting change in the firm's value due to productivity innovations. The time derivative of the value function accounts for the change of the aggregate state. The firm's innovation decision $\mu(z, t)$ is such that the marginal cost is equal to the marginal benefit from productivity improvement:

$$\Pi(t) c'(\mu(z, t)) e^z = V_z(z, t). \quad (11)$$

In equation (10), the right-hand side computes the flow payoff for imitators: they retain $1 - \beta$ share of the entry surplus, adjusting for the contact rate $\iota(\alpha(t))$.

It follows naturally from equation (9) that $V(z, t)$ is increasing in z . As such, there is a threshold $\underline{z}(t)$ such that firms choose to be active if and only if $z \geq \underline{z}(t)$. The innovation threshold $\underline{z}(t)$ satisfies the following value-matching and smooth-pasting conditions:

$$V(\underline{z}(t), t) = W(t) \quad (12)$$

$$V_z(\underline{z}(t), t) = 0. \quad (13)$$

3.3 Equilibrium Definition

The distribution of firm productivity evolves over time according to the following Kolmogorov Forward (KF) equation that satisfies, for all $z > \underline{z}(t)$,

$$\begin{aligned} f_t(z, t) = & \lambda(\alpha(t)) \left\{ \delta \int_0^\infty [f(z - q, t) - f(z, t)] dG(q) + (1 - \delta) \int_{\underline{z}}^\infty f(z - q, t) dG(q) \right\} \\ & - \frac{\partial}{\partial z} (\mu(z, t) f(z, t)) + \frac{1}{2} \sigma^2 f_{zz}(z, t), \end{aligned} \quad (14)$$

and, for all $z \leq \underline{z}(t)$, $f(z, t) = 0$.

The distribution of innovator productivity is shaped by forces of innovation and knowledge diffusion. On the one hand, there is an inflow of firms from the left to the right of the innovation threshold, as these firms acquire superior technology and jump ahead in the knowledge scale. At the same time, there is an outflow of innovating firms due to creative destruction associated with knowledge transfers. These two forces are captured

by the first term on the right-hand side of equation (14), which adjusts for the contact rate $\lambda(\alpha(t))$ and the creative destruction probability δ . On the other hand, the innovating firms are advancing along the knowledge scale at the speed of their innovation intensity $\mu(z, t)$ with a variance of σ . This force is captured by the last two terms.

We now define the equilibrium.

Definition 1 (Equilibrium). An equilibrium consists of value functions $\{S^e(z, t), V(z, t), W(t)\}$, innovation decision rule $\mu(z, t)$, innovation threshold $\underline{z}(t)$, fraction of innovators $\alpha(t)$, and productivity distribution $F(z, t)$, given the initial productivity distribution $F(z, 0)$, such that equations (8) to (14) are satisfied.

4 Balanced Growth Path

We focus the remainder of the analysis on the balanced growth path (BGP). Before proceeding to characterize the BGP, we first examine the investment pattern in the cross section among the incumbent innovators. We observe that according to the optimality condition (11) the innovation intensity must be proportional to the marginal firm value of improving productivity $V_z(z, t)$ relative to e^z . We further observe that an innovator firm faces an optimal stopping problem as captured by equations (12) and (13): the further away the innovator is from the threshold $\underline{z}(t)$, the longer it is expected to benefit from improved productivity before switching to the side of imitators. Therefore, the innovation intensity $\mu(z, t)$ increases with the innovator's productivity z . In the limit, for firms at the furthest point on the knowledge frontier, i.e., $z \rightarrow \infty$, the likelihood of exiting vanishes, so the innovation intensity converges to an upper bound $\mu(z, t) \rightarrow \bar{\mu}(t)$. We formally establish this pattern in Lemma 3 in Appendix A.4.

Along the BGP, the economy grows at a constant rate $g(t) = g$. The fraction of innovator firms is also constant $\alpha(t) = \alpha$. The value functions of innovators and imitators satisfy $V(z + (\eta - 1)gt, t) = e^{gt}V(z)$ and $W(t) = e^{gt}W$; the productivity distribution can be transformed into a stationary one,

$$f(z + (\eta - 1)gt, t) = f(z). \quad (15)$$

Further, the innovation threshold grows at rate $(\eta - 1)g$, $\underline{z}(t) = e^{(\eta-1)gt}\underline{z}$. Finally, the innovation upper bound also converges to a constant, $\bar{\mu}(t) = \bar{\mu}$.

Assumption 1. *The initial productivity distribution $F(z, 0)$ has a bounded support.*

Under this assumption, the initial set of knowledge in the economy is bounded and limited. Therefore, the creation of new knowledge through innovation is essential to generate sustained economic growth. Otherwise, in the absence of any innovative effort there cannot be sustained growth. This assumption is a sufficient condition to ensure the existence of a unique BGP, as in [Luttmer \(2007\)](#).⁹

The following proposition characterizes the BGP.

Proposition 2 (BGP). *Under Assumption 1, there exists a unique BGP with the following features.*

(i) Along the BGP, the economy grows at rate g given by

$$g = \frac{1}{\eta - 1} \left\{ \underbrace{\bar{\mu} + \frac{1}{2}\sigma^2\zeta}_{\text{own innovation}} + \underbrace{\lambda(\alpha) D(\zeta)}_{\text{knowledge diffusion}} \right\}. \quad (16)$$

where

$$D(\zeta) = \frac{1}{\zeta} \left[\underbrace{\delta \int_0^\infty (e^{\zeta q} - 1) dG(q)}_{\text{creative destruction}} + \underbrace{(1 - \delta) \int_{\underline{q}}^\infty e^{\zeta q} dG(q)}_{\text{new varieties}} \right].$$

(ii) The upper bound of innovation $\bar{\mu}$ is characterized by¹⁰

$$c'(\bar{\mu}) = \frac{1 - c(\bar{\mu})}{\rho + (\eta - 1)g - \bar{\mu} - \frac{1}{2}\sigma^2 - \lambda(\alpha)(\beta s^e - \hat{\delta})}. \quad (17)$$

(iii) The endogenous stationary distribution has an asymptotic Pareto tail with parameter ζ , which satisfies the following equation:

$$\frac{1}{2}\sigma^2 + \lambda(\alpha) D'(\zeta) = 0. \quad (18)$$

⁹If the initial productivity distribution is unbounded, there can potentially exist a continuum of equilibria associated with different growth rates and stationary distributions. For example, [Benhabib et al. \(2021\)](#) distinguishes between a unique growth path and hysteresis, depending on the initial productivity distribution.

¹⁰The preference and technology parameters satisfy $\rho > \beta \left(\int e^q dG(q) - \delta \right) + \frac{1}{2}\sigma^2 - g$. The agents discount future consumption at a sufficiently high rate. This assumption is merely to ensure that, along the BGP, the firm's value functions are bounded and hence well-defined.

The first part of Proposition 2 provides a formula for the growth rate. The terms inside the bracket of Equation (16) capture the shift in the distribution of productivity which is then translated to growth in output dividing by $\eta - 1$. The expression decomposes the growth rate into two sources: the speed of innovation by the firms on the knowledge frontier, $\bar{\mu} + \frac{1}{2}\sigma^2\zeta$, and the impact of knowledge diffusion, $\lambda(\alpha)D(\zeta)$. In terms of knowledge diffusion, $\lambda(\alpha)$ is the intensity at which existing technologies are diffused (the contact rate), and the function $D(\zeta)$, which depends on the tail parameter ζ , summarizes the knowledge diffusion force. The knowledge diffusion function $D(\zeta)$ can be further decomposed into two parts: When an imitator and innovator match, creative destruction occurs with a probability of δ , and a new variety is created with a probability of $1 - \delta$. Similar formulas are obtained in continuous-time models in previous papers by Luttmer (2007, 2012) and Benhabib et al. (2021). Note that all components in the formula are endogenous outcomes driven by the incentives for innovation and imitation, including the fraction of firms innovating α , the associated matching rate $\lambda(\alpha)$, and the innovation intensity $\bar{\mu}$. The relative strength of the bargaining power between the innovators and the imitators affects their incentives and therefore trades off the two sources of growth.

The last part of Proposition 2 characterizes the detrended stationary productivity distribution. This distribution does not have an analytical solution because the innovation intensity $\mu(z)$ varies with the firm's distance to the innovation threshold. However, on the knowledge frontier, as the innovation intensity $\mu(z)$ converges to its constant upper bound $\bar{\mu}$, we can characterize the distribution in the right tail: it follows an asymptotic Pareto distribution with a shape parameter in equation (18), a useful measure of productivity heterogeneity. The variance in innovation outcome and the diffusion rate together determine the shape of productivity distribution in the right tail. A higher standard deviation σ leads to a fatter right tail, while a higher diffusion rate $\lambda(\alpha)$ leads to a thinner right tail.

Innovation and knowledge diffusion are two opposing forces in shaping the extent of equilibrium heterogeneity. On the one hand, innovation contributes to firm heterogeneity, stretching out the productivity distribution. The dispersion is mainly due to the stochastic nature of the innovation outcome, as captured by the standard deviation of the Brownian motion. Knowledge diffusion, on the other hand, compresses the productivity distribution as the less productive firms are matched with the more productive incumbents. As seen earlier, the speed of renewal is very closely related to innovator's contact rate, which explains why increases in $\lambda(\alpha)$ decrease productivity dispersion.

Growth in the entire distribution. Another growth accounting can be obtained by looking at the entire sample of active firms across the whole distribution, i.e., for all

$z \geq \underline{z}$. To compute the aggregate productivity growth, we again look at the three types of innovation and knowledge diffusion events one by one.

We start with own-innovation activities. Modifying the previous growth formula (37) for the tail, we now account for varying levels of innovation intensity across the distribution. For a firm with productivity z choosing innovation intensity $\mu(z)$, its expected growth is $\mu(z) + \frac{1}{2}\sigma^2$. The growth due to own innovation is the weighted average of the individual firm growth rates. Next, knowledge diffusion events of the creative destructive nature occur at rate $\lambda(\alpha)\delta$ and lead to expected productivity improvement of $\int_0^\infty (e^q - 1) dG(q)$. Similarly, knowledge diffusion events generating new varieties arrive at rate $\lambda(\alpha)(1 - \delta)$. The new varieties have an expected productivity level $\int_{\underline{q}}^\infty e^q dG(q)$ relative to the average incumbent firms. Finally, with the arrival of each new variety, there will be a corresponding exit of an existing variety to balance the composition of incumbents and potential entrants. Therefore, the exit rate is equal to the new variety arrival rate. The exiting firms are the marginal firms at the threshold $\underline{z}$; therefore, their impact on the growth depends on how large they are relative to the average incumbent, i.e., the *margin-to-mean ratio* $e^{\underline{z}}/\mathbb{E}[e^z]$. Putting these forces together, we obtain the aggregate productivity growth rate:

$$g = \frac{1}{\eta - 1} \left\{ \underbrace{\frac{\mathbb{E}[e^z \mu(z)]}{\mathbb{E}[e^z]} + \frac{1}{2}\sigma^2}_{\text{own innovation}} + \underbrace{\lambda(\alpha)\delta \int_0^\infty (e^q - 1) dG(q)}_{\text{creative destruction}} + \underbrace{\lambda(\alpha)(1 - \delta) \int_{\underline{q}}^\infty e^q dG(q)}_{\text{new varieties}} - \underbrace{\lambda(\alpha)(1 - \delta) \frac{e^{\underline{z}}}{\mathbb{E}[e^z]}}_{\text{exit}} \right\}. \quad (19)$$

The margin-to-mean ratio $e^{\underline{z}}/\mathbb{E}[e^z]$ in the exit component of equation (19) captures the extent of incumbent productivity heterogeneity. This measure encompasses the equilibrium level of knowledge diffusion, the dispersion in innovation outcomes, and the variation in the innovation efforts. This ratio adjusts such that the two growth accounting are aligned, as we show in the following lemma. This insight is more clearly conveyed in an exogenous constant innovation benchmark, $\mu(z) = \bar{\mu}$, in Online Appendix A.8, a case closed forms can be obtained for the whole distribution.

Lemma 1. *The growth rate in equation (19) coincides with the growth formula (16) in the tail.*

4.1 “Lucks” and Heterogeneity

Remember that if the productivity of the matched innovator is z , then the new productivity of the imitator is $z + q$, where q is drawn from an exogenous distribution $G(\cdot)$. Here, we consider the special case where $q = 0$: an imitator always perfectly imitates an innovator.

Lemma 2 (Perfect and Deterministic Imitation). *Suppose that the imitation is perfect and deterministic, i.e.,*

$$G(q) = \begin{cases} 0 & q < 0 \\ 1 & q \geq 0 \end{cases}.$$

Along the BGP, the endogenous stationary distribution has a Pareto tail index

$$\zeta = \frac{\sqrt{2\lambda(\alpha)(1-\delta)}}{\sigma}. \quad (20)$$

and the economy grows at rate

$$g = \frac{1}{\eta - 1} (\bar{\mu} + \sigma^2 \zeta), \quad (21)$$

When imitation is perfect and deterministic, the productivity of imitators is always the same as that of innovators and never exceeds it, so creative destruction that occurs with probability δ always ends in failure. Therefore, the productivity distribution changes only when a new variety is born with probability $1 - \delta$, and the effective matching rate is $\lambda(\alpha)(1 - \delta)$. The tail parameter ζ is expressed in terms of the matching rate and the variance of own innovation, as shown in previous models in the literature (e.g. [Luttmer \(2015\)](#)). As the matching rate $\lambda(\alpha)(1 - \delta)$ increases, the tail becomes thicker. As the variance of own innovation becomes larger, the tail becomes thinner.

The growth rate is expressed as the sum of the drift of active firms in the tail $\bar{\mu}$ and the product of the variance and the tail parameter $\sigma^2 \zeta$, which is also an analogous result to [Luttmer \(2015\)](#). Following Proposition 2, we can decompose the factors of growth into own innovation and knowledge diffusion:

$$g = \frac{1}{\eta - 1} \left\{ \underbrace{\bar{\mu} + \frac{1}{2}\sigma^2\zeta}_{\text{own innovation}} + \underbrace{\frac{1}{2}\sigma^2\zeta}_{\text{knowledge diffusion}} \right\}$$

As we can see from this growth accounting, the large dispersion of innovation σ contributes to economic growth through both own-innovation and knowledge diffusion.

4.2 Innovator Appropriation and BGP

The bargaining power between innovators and imitators shapes firms' incentives to innovate or imitate. We now study how changes in bargaining weights affect the equilibrium composition and growth dynamics of the economy.

As the bargaining position of innovators strengthens, a larger proportion of firms choose to innovate rather than imitate. Consequently, the diffusion rate $\lambda(\alpha)$ goes down. Stronger innovator appropriability reduces the speed of knowledge diffusion and, hence, its contribution to aggregate growth.

The innovation intensity is driven by the direct and indirect effects from changes in the bargaining weights. It is useful to observe the first-order condition (17) that characterizes the upper bound of innovation $\bar{\mu}$. As the bargaining position of innovators strengthens, directly, the innovator firms appropriate more rents from knowledge diffusion, which induces a higher innovation effort. At the same time, an indirect force due to congestion offsets that innovation incentive. As noted above, as β increases, α increases and $\lambda(\alpha)$ goes down. This negative indirect effect could dominate the positive direct effect. Combining the two effects, the indirect payoff to innovation from knowledge transfer $\beta\lambda(\alpha)$ might exhibit a hump shape in the innovator's bargaining weight β . Finally, there is an additional equilibrium effect that needs to be considered. Higher diffusion increases the growth rate of the economy and thus the rate at which knowledge depreciates over time. As can be seen in equation (17), this has an adverse effect on innovation.

The overall effect of the bargaining weights on the growth rate is also likely non-monotonic. In the lower range of the innovator's bargaining weight, a higher growth through innovation could offset the decline in the growth due to knowledge diffusion.

We further characterize the limiting behavior of the economy as innovators appropriate almost the full rent from knowledge diffusion. As enforcement strengthens and the innovators' effective bargaining weight approaches one, leaving imitators with almost none of the surplus, the equilibrium composition of innovators and imitators adjusts to restore the balance of incentives between innovation and imitation. That is, for the indifference condition (12) to hold, the expected surplus from knowledge diffusion must remain well-defined. This requires the expected surplus to be integrable, i.e., $\zeta > 1$, and the endogenous diffusion rate $\lambda(\alpha)$ to adjust above a minimum level. We summarize this limit below.

Corollary 1 (Limiting Economy). *As enforcement $\beta \rightarrow 1$, the fraction of innovators $\alpha \rightarrow \alpha^1$, which satisfies $\lambda(\alpha^1) = -\frac{1}{2}\frac{\sigma^2}{D'(1)}$. The tail thickens $\zeta \rightarrow 1$, and the growth rate $g \rightarrow \frac{1}{\eta-1} \left(\bar{\mu}^1 + \frac{1}{2}\sigma^2 \left(1 - \frac{D(1)}{D'(1)} \right) \right)$.*

Table 3: Calibrated parameters

Parameter	Value	Moment	Data
<i>Standard</i>			
Discount rate ρ	0.032	Annual interest rate	5%
Elasticity of substitution η	4	Preset	
<i>Innovation process</i>			
Standard deviation σ	0.11	Own innovation randomness (GHK)	0.11
Cost function scale χ	629	Own innovation mean (GHK)	1.15%
Cost function elasticity ε	1	Preset	
<i>Diffusion process</i>			
Matching function curvature ω	0.5	Preset	
Matching function efficiency κ	4.8	Total entry rate (GHK)	7.0%
Creative destruction rate m_{CD}	5.4%	Creative destruction rate (GHK)	5.4%
New variety rate m_{NV}	1.6%	New variety rate (GHK)	1.6%
Quality draw shape ϕ	4.87	Quality draw shape (GHK)	4.87
Quality draw lower bound $\underline{q}$	-1.37	New variety average quality (GHK)	0.32
Innovator bargaining weight β	0.5	Royalty rate (Kankanhalli Kwan 2022)	

Notes: The creative-destruction and new-variety entries are annual arrival rates from GHK. The model maps these rates to the primitive branch probability δ and the effective displacement probability $\hat{\delta} = \delta(1 - G(0))$.

Through the lens of the perfect and deterministic imitation example in Lemma 2, $D(1) = 1$ and $D'(1) = -1$, the growth rate from imitation $\lambda(\alpha)(1 - \delta)$ converges to $\frac{1}{2}\sigma^2$, and the overall growth rate converges to $\frac{1}{\eta-1}(\bar{\mu} + \sigma^2)$.

This limiting case also reveals that the economy exhibits a discontinuity when innovators hold all the bargaining power. In this extreme case with $\beta = 1$, no firm would want to be on the imitation side, and knowledge diffusion completely disappears, i.e., $\alpha = 1$ and $\lambda(\alpha) = 0$. The growth comes solely from the innovators and reduces to $\frac{1}{\eta-1}(\bar{\mu} + \frac{1}{2}\sigma^2)$.

5 Quantitative Evaluation

In this section, we first calibrate the model and then quantitatively assess the impact of knowledge appropriability on the overall economic growth.

5.1 Calibration

To calibrate the model, we borrow the estimate from Garcia-Macia et al. (2019) (henceforth GHK) regarding the extent of creative destruction of innovation as opposed to own innovation and the arrival of new varieties based on the patterns of job creation and job

destruction at the firm level during the period 1983-2013.¹¹

We assume that the economy is on a BGP and take advantage of the properties of the economy along the BGP in Proposition 2 to map the parameters to their corresponding moments. The calibration is carried out at annual frequency: one unit of time in the model corresponds to one year in the data. Table 3 reports the calibrated parameter values. To match an annual interest rate of around 5% and an overall growth rate of 1.76%, we set the discount rate ρ to 0.032 according to the interest rate equation $r = \rho + g$. Following the standard in the literature, we set the elasticity of substitution parameter η to 4.

Standard deviation. The standard deviation of the Brownian motion σ corresponds to the volatility of the growth rate of the largest firms in the economy.¹² We instead calibrate the innovation and diffusion process directly from GHK so that expected firm growth due to own innovation satisfies $\bar{\mu} + \frac{1}{2}\sigma^2 = \lambda_i \frac{1}{\phi-1}$ and the variance of firm growth is $\sigma^2 = \frac{2\lambda_i}{\phi^2}$, where $\mathbb{E}[\lambda_i] = 0.146$.

Innovation cost function. The innovation function is specified as $c(\mu) = \frac{\chi}{\varepsilon+1}\mu^{\varepsilon+1}$. We fix the elasticity parameter ε at 1, following the macro and micro estimates of innovation elasticity around unity in the literature. We calibrate the innovation cost scale parameter χ to match the contribution of own innovation to growth, $\bar{\mu} + \frac{1}{2}\sigma^2\zeta = 1.15\%$, and obtain $\chi = 629$.

Quality draw. For the quality draw $G(q)$, we specify the exponential distribution in Example 1. First, consider the knowledge diffusion events that lead to creative destruction. The conditional distribution for the quality draw $\tilde{G}(q)$ has the same shape as $G(q)$ after being truncated for $q \geq 0$. This distribution is identical to one specified in Garcia-Macia et al. (2019). Therefore, we map the shape parameter to one they calibrated, $(\eta - 1)\phi = 14.6$, and obtain that $\phi = 4.87$. Next, consider the knowledge diffusion events that lead to new varieties. To match an average quality of new varieties equal to 32% of the incumbents' level, we infer that the lower bound is $\underline{q} = -1.37$.

Creative destruction and new varieties. To gauge the extent of creative destruction, we also borrow the estimates from GHK. They compute that, on an annual basis, the

¹¹The firms in our model are single-product, whereas GHK focus on multi-product firms, so their notion of firms or establishments versus products differs from ours.

¹²Using the Longitudinal Business Database, Davis, Haltiwanger, Jarmin and Miranda (2007) find that the volatility of the growth rate for large firms is in the range of 0.05 to 0.1. This measure of the volatility of firm growth rate in Davis et al. (2007) excludes short-lived firms and entry and exit.

arrival rate of creative destruction is 0.054, and the arrival rate of new varieties is 0.016. Decomposing into entrants and incumbents, the creative destruction rate by entrants is 0.0345 and the rate by incumbents is 0.0195. Our model maps to the GHK environment except for one discrepancy: in contrast to their multi-product firms, the firms in our model are single-product firms and an incumbent does not take over a variety owned by another firm. Thus, creative destruction occurs only through entrants. For the purpose of our analysis, we lump creative destruction by incumbents into creative destruction by entrants to capture sources of knowledge diffusion.¹³

Let $m_{CD} = 0.054$ denote the annual arrival rate of creative destruction and $m_{NV} = 0.016$ the annual arrival rate of new varieties. These are rate targets, not primitive probabilities. In the model they satisfy

$$m_{CD} = \lambda(\alpha) \delta (1 - G(0)), \quad m_{NV} = \lambda(\alpha) (1 - \delta).$$

Given the calibrated quality draw distribution, the share of creative-destruction draws that can displace the incumbent is $1 - G(0) = 0.0013$. The two rate restrictions imply $\delta = 0.9996$ and $\hat{\delta} \equiv \delta (1 - G(0)) = 0.00127$. The ratio $s_{CD} = m_{CD}/(m_{CD} + m_{NV}) = 0.77$ summarizes the GHK decomposition of observed diffusion events, but it is not a primitive model probability.

Bargaining power. Patent enforcement—and the corresponding bargaining power—is a parameter that we know little about. Several recent empirical studies help fill this gap. Using licensing transactions for US public firms, [Kankanhalli and Kwan \(Forthcoming\)](#) finds that the royalty rate on average amounts to 11.5% of net sales, or around a half-and-half split of the profits. Therefore, we set β to 0.5, assigning a half-and-half split of entrant value to the transacting parties.¹⁴

To gauge whether this calibrated bargaining weight is sensible, we cross-check a few model moments directly related to the bargaining weights. First, the bargaining weight affects a firm's indirect payoff from innovation in addition to direct profit from production. Our calibrated model implies that licensing contributes to around 12.8% of total revenue for the largest firms.¹⁵ Second, the bargaining weight β shapes the private return of

¹³Another interpretation of our results is learning at the product level, and firms are a collection of products.

¹⁴Practitioners in licensing transactions commonly refer to a “25% rule”, which states that the licensees pay a 25% royalty rate out of the profits from the licensing deals to the licensors.

¹⁵For the largest firms, the licensing income is $\beta \lambda(\alpha) D \bar{V}$, and the direct revenue is $\eta \pi (1 - c(\bar{\mu}))$. The share $\frac{\beta \lambda(\alpha) D \bar{V}}{\eta \pi (1 - c(\bar{\mu})) + \beta \lambda(\alpha) D \bar{V}} = 12.8\%$.

innovation relative to the social return. When the innovators appropriate all gains from knowledge diffusion, the gap between the private return and the social return closes to zero. Bloom et al. (2013) estimate that the social rate of return to R&D is around 55%, while the private return is 21%. While our model doesn't directly map to their calculation, we have a notion of social value versus private value, which has a ratio of 2.5.¹⁶

Matching function. We work with the Cobb-Douglas matching function: $M(\alpha, 1 - \alpha) = \kappa \alpha^\omega (1 - \alpha)^{1-\omega}$. The corresponding contact rate for innovators is $\lambda(\alpha) = \kappa \left(\frac{\alpha}{1-\alpha}\right)^{\omega-1}$ and for imitators $\iota(\alpha) = \kappa \left(\frac{\alpha}{1-\alpha}\right)^\omega$. The level parameter κ captures the ease of knowledge diffusion. The matching elasticity $\omega \in [0, 1]$ controls the contribution of innovators to matching.

Although this specification of the matching function is widely examined in studies of labor markets, it is less explored in studies related to ours for technology markets. One exception is Akcigit, Celik and Greenwood (2016) who look into the patent resale market through a random search model. They also specify a Cobb-Douglas matching function and, using the empirical distribution of the duration until a patent gets sold, find an equal amount of congestion on the two sides of the market. Hence, we set ω to 0.5 such that the extent of congestion is equal on the two sides. The parameter κ is calibrated to match the contribution of knowledge diffusion to growth. To target the new variety rate $\lambda(\alpha)(1 - \delta) = 0.016$, we set κ to 14. The model-implied fraction of firms on the innovating side is roughly 0.9%. While the magnitude of this share is not directly interpretable, it is helpful to view the imitator side as a large number of potential entrants.

5.2 Growth Decomposition

We decompose the growth rate into different sources using the calibrated model. We do so both for firms in the tail according to equation (16) and for firms across the whole distribution according to equation (19).

Overall, the growth decomposition in the tail is more or less identical to the one across the whole distribution. Own innovation accounts for 2/3 of overall growth and diffusion accounts for the remaining 1/3. The two decompositions indeed sum to the same total growth rate, as stated in Lemma 1. The small discrepancies in the decomposed levels stem from cross-sectional heterogeneity in the calibrated model. In particular, the growth due to own innovation in the tail is always higher than the level that averages over the

¹⁶The private value of R&D is the right-hand side of equation (17), while the social value is obtained by setting $\beta = 1$ in the same expression.

entire distribution, i.e., $\bar{\mu} + \frac{1}{2}\sigma^2\zeta > \mathbb{E}[\mu(z)e^z] / \mathbb{E}[e^z] + \frac{1}{2}\sigma^2$. The gap remains modest for two reasons. First, the Pareto tail index ζ implied by equation (18) is 1.17, close to the empirical benchmark of about 1.1 in the U.S. data. Second, dispersion in innovation intensity is limited. Consistent with this observation, the margin-to-mean ratio for new varieties, $e^z / \mathbb{E}[e^z]$, is only 0.019, indicating that entrants begin far from the tail and add little dispersion.

Counterfactuals: Appropriability and Growth We evaluate how the extent of appropriation in knowledge diffusion affects the incentives for innovation and imitation and hence the overall economic growth. We use the calibrated model to carry out the counterfactual exercise by varying the bargaining weights. Figure 1 plots, as the innovator's bargaining weight β increases from 0 to 1, the corresponding growth rate and its decomposition into innovation and knowledge diffusion.

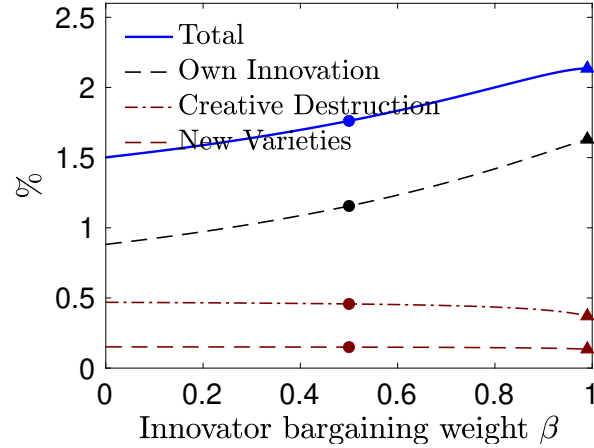
The growth rate is maximized when the innovator's bargaining weight is 0.86. The optimal bargaining weight trades off two opposing forces for growth. On the one hand, it discourages knowledge diffusion as imitation becomes less profitable. On the other hand, it encourages innovation, as the innovators appropriate a larger share of the surplus created by knowledge transfer.¹⁷ We find that the first effect is very mild since, as β is increased from 0 to 0.86, the fraction of imitators falls very slowly from 58% to 47%. This in turn explains why the contribution of knowledge diffusion to growth depicted in Figure 1 is so flat as a function of β . On the other hand, the elasticity of innovation to the innovator's bargaining weight is large, as the innovator's return from licensing represents 20% of their total revenue at the calibrated level and it varies quite substantially with the innovator's appropriability. The indirect payoff from licensing increases from 0% to 47% of total revenue as β is increased from 0 to 0.86. This explains the fairly steep rise in the contribution of innovation to growth observed in Figure 1. Overall, the long-run growth rate increases from the calibrated value of 2% when $\beta = 0.3$, the benchmark level, to 2.31% for the optimal value $\beta = 0.86$. It should be noted that while this degree of optimal enforcement might seem high, the net share of the value appropriated by innovators, $(\beta^* - \delta) / (1 - \delta)$, is much lower, representing 0.58 of the net surplus created.

5.3 Ramsey Problem

In this section, we analyze the optimal bargaining power in our dynamic model. A Ramsey planner chooses the stream of bargaining power $\{\beta(t)\}_{t \geq 0}$. By doing so, she influences

¹⁷At very high levels of β there is a Laffer curve as the share of imitators vanishes.

Figure 1: Bargaining weight and growth rate



Notes: Growth rate and its innovation and diffusion components as the innovator's bargaining weight β varies from 0 to 1. Growth peaks at $\beta = 0.86$, where the rate reaches 2.31%.

firms' choices so as to maximize household utility. We focus on the Ramsey steady state that emerges as a result of the stream of optimal bargaining power.

For tractability, we treat the innovation intensity μ as exogenous and, following Section 4.1, assume imitators perfectly replicate the incumbents' productivity so that $q = 0$ at each diffusion event. Under this assumption, we can characterize the optimal bargaining power analytically. Once the share of innovators α in the steady-state equilibrium is determined, the steady state is uniquely pinned down. Therefore, we follow the steps below to obtain the optimal bargaining power β in the Ramsey steady state. (i) First, we solve the benevolent social planner's problem, which maximizes household utility subject to the KF equation (14), and compute the steady-state value of α . (ii) Next, we identify the value of β that generates this α in the competitive steady state.

The following proposition characterizes the Ramsey steady state when the innovation intensity μ is exogenous.

Proposition 3 (Ramsey Steady State with Exogenous Innovation). *When the innovation intensity μ is exogenous and imitators perfectly replicate the incumbents' productivity ($q = 0$), the optimal bargaining power β and the share of innovators α in the Ramsey steady state are*

characterized by the following equations:

$$(\rho - \lambda(\alpha)(1 - \delta)) \alpha \left(1 + \frac{1}{\tilde{v}}\right) = \psi(\alpha)(1 - \delta)(1 - \omega) \left(\left(\frac{\zeta}{\zeta - 1}\right)^2 - 1 + \frac{1}{\tilde{v}} \left(\left(\frac{\zeta}{\zeta + \tilde{v}}\right)^2 - 1 \right) \right), \quad (22)$$

$$(\rho - \lambda(\alpha)(1 - \delta)\beta) \left(1 + \frac{1}{v}\right) = \psi(\alpha)(1 - \delta)(1 - \beta) \left(\left(\frac{\zeta}{\zeta - 1}\right)^2 - 1 + \frac{1}{v} \left(\left(\frac{\zeta}{\zeta + v}\right)^2 - 1 \right) \right), \quad (23)$$

where ζ is pinned down by (20). The term $\tilde{v}$ in the planner's problem equation (22) and the term v in the competitive equilibrium equation (23) satisfy

$$\begin{aligned} \left(\frac{\zeta}{\zeta + \tilde{v}}\right)^2 &= \frac{\lambda(\alpha)(1 - \delta)}{\rho}, \\ \left(\frac{\zeta}{\zeta + v}\right)^2 &= \frac{\lambda(\alpha)(1 - \delta)}{\rho + \lambda(\alpha)(1 - \delta)(1 - \beta)}. \end{aligned}$$

Under exogenous μ , the steady-state share of innovators α in the benevolent planner's allocation of the dynamic model is characterized by equation (22). The relationship between bargaining power β and the share of innovators α in the competitive steady state is given by equation (23). Therefore, substituting the α obtained from (22) into (23) yields the optimal bargaining power β .

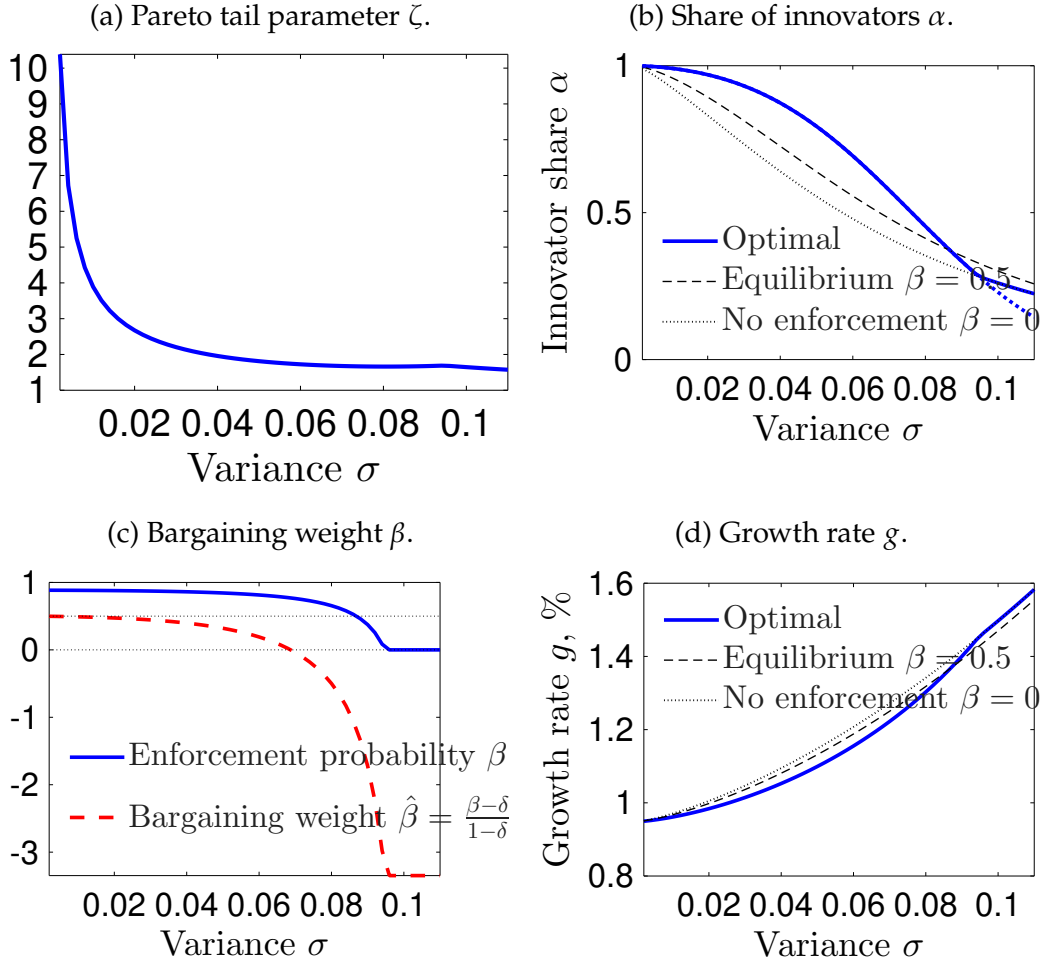
In implementing this Ramsey problem with an exogenous innovation rate, we retain the parameter values from the dynamic calibration in Section 5.1. The exogenous innovation intensity is set to $\mu = 0.0285$ so that the contribution of own innovation to growth matches the 1.15% estimate reported by Garcia-Macia et al. (2019).

As shown in Figure 2a, greater dispersion in firm productivity raises both the variance across firms and the Pareto tail parameter ζ . A thicker tail makes learning appealing to more firms, so the share of innovators α drops (Figure 2b). When heterogeneity is large, the displacement effect is strong, as discussed in Section 2.2, and the planner gives innovators less bargaining power and lowers β (Figure 2c).

When σ is high, the optimal enforcement probability falls below the baseline that assumes $\beta = 0.5$ (Figure 2c). In that range, the share of innovators falls under the baseline (Figure 2b). Faster technology diffusion in the Ramsey steady state then raises the growth rate above the benchmark outcome (Figure 2d).

Figure 3 shows the optimal bargaining weight for different matching elasticities ω . As in Section 2.2, a higher matching elasticity ω magnifies the congestion externality on the

Figure 2: Planner allocations under the Ramsey steady state



Notes: Panels display the planner solutions for the Pareto tail, the share of innovators, the bargaining weight, and the growth rate, respectively. The x-axis reports the dispersion of firm productivity, σ , so each panel compares the allocation under various variances.

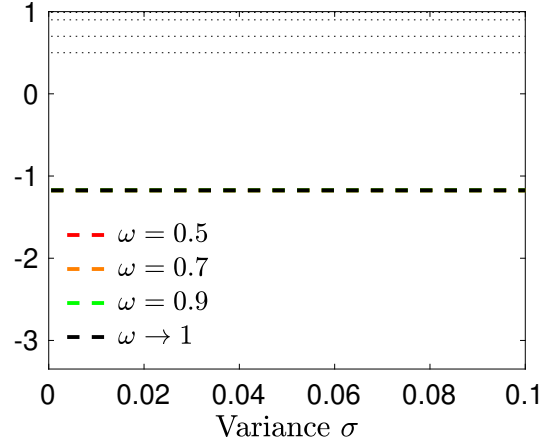
imitator side, so the planner grants innovators more bargaining power and raises β . In the dynamic model, the [Hosios \(1990\)](#) condition $\beta = \omega$ does not hold exactly, and the gap widens as firm heterogeneity σ increases.

6 Extensions

6.1 Scalable-Volatility Innovation Process

The counterfactual analysis above reveals that the endogenous level of innovator heterogeneity is critical for determining the optimal level of enforcement. One important factor

Figure 3: Bargaining weights across matching elasticity



Notes: The figure plots the Ramsey planner's optimal bargaining weight $\hat{\beta}$ as the curvature of the matching function, ω , varies. The x-axis reports σ , and the lines compare allocations across alternative levels of matching elasticity, ω .

driving this heterogeneity is the dispersion of innovation outcomes. In the model in Section 3, regardless of the innovation effort, the variance of the outcome is fixed. We call this a *fixed-volatility* productivity process, represented by a Brownian motion with an upward drift μ and a standard deviation σ . This model of innovation is commonly used in the literature, for example, in the continuous-time model of Luttmer (2007).

We contemplate an alternative setting in which a firm's endogenous innovation decision can affect not only the expected level but also the variance of productivity improvement. We capture the effect on the variance with a flexible specification of the Brownian motion, which has a standard deviation $\sigma(\mu)$ as a function of innovation intensity μ . Specifically, we introduce a *scalable-volatility* innovation process, represented by a Brownian motion with standard deviation $\sigma(\mu) = \sigma\sqrt{\mu}$. That is, firm productivity follows

$$dz = \mu dt + \sigma\sqrt{\mu}dB. \quad (24)$$

To micro-found this scalable-volatility productivity process, we consider firm innovation as experimentation through a set of risky projects. Let μ denote the number of projects carried out at a given point in time. Each project generates some incremental productivity improvement with an outcome drawn independently from a normal distribution $N(1, \sigma^2)$. Together the μ projects generate productivity improvement drawn from a normal distribution $N(\mu, \sigma^2\mu)$. Firm innovation follows a quality ladder in which each discovery raises productivity by one rung.

A distinctive feature of the scalable-volatility productivity process is that there is no free luck in innovative discoveries. If firms expend more resources in innovation and carry out more experimentation, they are more likely to be lucky in discovering better productive processes, leading to more dispersed outcomes. The level of innovation intensity affects the variance of innovation outcomes and in turn the equilibrium extent of firm heterogeneity.

We calibrate the dynamic model with the scalable-volatility productivity process, following the same strategy as with the fixed-volatility model. Only three parameters are altered here. First, the growth decomposition into innovation and knowledge diffusion is unchanged. After obtaining the extent of innovation intensity from the decomposition, to fit the innovating firm's volatility $\sigma\sqrt{\bar{\mu}}$ to 0.08, we back out σ as 0.94. Second, since the optimality condition for innovation is altered now to account for the additional effect of a higher volatility on the firm value, to target the innovation intensity, we set the innovation-cost parameter γ to 4100. Third, to target the knowledge diffusion rate, we set ϕ to 0.038. The model-implied fraction of innovating firms is 49%.

We then carry out the counterfactual exercise of the growth-maximizing level of innovator appropriation. As the innovator's bargaining weight β increases from 0 to 1, the corresponding growth rate and its decomposition into innovation and diffusion. Compared to the fixed-volatility model, here the overall growth rate tends to be maximized by an even higher level of innovator bargaining weight, $\beta^* = 0.9$. The corresponding level of net appropriation by innovators, $(\beta^* - \delta) / (1 - \delta)$, is 0.7. The scope of improving long-run growth is larger with scalable volatility: the growth rate increases from the calibrated value of 2% when $\beta = 0.3$ to 2.66% at the optimal value $\beta^* = 0.9$.

One distinctive feature of the scalable-volatility innovation process is how knowledge diffusion responds to the extent of innovator appropriation. With fixed volatility, growth through knowledge diffusion decreases monotonically as the innovator's bargaining position strengthens. In contrast, with scalable volatility, for a large interval of the innovator's bargaining weight, the diffusion rate increases as the innovator's bargaining weight strengthens. This is because, at a higher innovator appropriation, as the extent of innovator experimentation $\bar{\mu}$ increases, the dispersion $\sigma\sqrt{\bar{\mu}}$ also increases. This distinction can also be seen through the pattern of the Pareto tail index, which experiences a sharper decrease in the case with scalable volatility compared to the case with fixed volatility. As a result, the decline in the share of imitators is even milder: it falls from 49% to 44% as β increases from 0 to 0.86. The response of the innovation rate, on the other hand, is similar in the two cases. This is because the indirect payoff to innovation is largely driven by the change in appropriability in both cases. Here, the innovator's return from licensing

represents 19% of their total revenue at the calibrated level and it increases from 0% to 49% of total revenue as β is increased from 0 to 0.9.

7 Conclusion

Recent macro models of growth and diffusion, such as [Moll and Lucas \(2014\)](#) and [Perla and Tonetti \(2014\)](#), highlight how random matching spreads ideas. Our contribution is to bring three missing ingredients to that framework and quantify their interaction. First, we model matching frictions explicitly, allowing the ratio of innovators to imitators to shape contact rates. Second, we incorporate creative destruction so that imitators can displace innovators and erode their rents. Third, we let the bargaining powers of innovators and imitators span the full range of surplus splits, tying intellectual property enforcement to measured growth outcomes.

The static model delivers a modified Hosios condition for homogeneous firms, clarifying how congestion forces adjust the benchmark share innovators should appropriate. Once we allow for heterogeneity, two forces push the optimum away from that benchmark: innovators earn additional rents from heterogeneous productivity draws, and the marginal match becomes less valuable to imitators. Consequently, the optimal bargaining weight declines with dispersion and can fall to zero when heterogeneity is sufficiently pronounced.

These insights hold in our quantitative dynamic model. The calibrated model implies that growth is maximized at an innovator bargaining weight of $\beta^* = 0.86$. This raises the long-run growth rate from 2.00 to 2.31 percent. Net of creative destruction, this optimum corresponds to innovators appropriating about 58 percent of the surplus created through licensing. When we vary the matching elasticity, the optimal β that maximizes social welfare spans the full range from 0 to 1, underscoring how congestion in the matching market is pivotal for policy.

At the current stage, the dynamic environment keeps innovation intensity exogenous while we study optimal bargaining weights. Future work will endogenize the innovation effort and solve the resulting Ramsey problem. This will let us trace welfare implications and provide further guidance for intellectual property design.

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Appendix

A Derivations and Proofs

A.1 Proof of Proposition 1

TBA

A.2 The Extended Enforcement Game

In the one-period model presented in Section 2, we assumed for illustrative purposes that knowledge transfers always create new varieties, so the enforcement game that determines the surplus division is straightforward. In the dynamic model in Section 3, by contrast, a transfer may lead to creative destruction, allowing the imitator to displace the incumbent. This changes the outside options of both parties in the licensing stage.

For simplicity, we assume that the imitation outcome is realized before the enforcement game.¹⁸ Figure 4 depicts the agents' payoffs in the cases in which entry occurs. When the realized draw does not permit entry, the match leaves both parties at their status quo payoffs.

Condition on a meeting between an innovator with productivity z and an imitator at time t , and let the realized imitation outcome be q . Define

$$S(z, t) \equiv V(z, t) - W(t).$$

After the realization of q , the imitator's entrant surplus is therefore $S(z + q, t)$. Let t_L denote the license fee. The imitator accepts whenever

$$W(t) + S(z + q, t) - t_L \geq W(t) + (1 - \beta) S(z + q, t),$$

which is equivalent to $t_L \leq \beta S(z + q, t)$. Since the innovator's payoff under licensing is increasing in t_L , it sets $t_L = \beta S(z + q, t)$. When the imitator is indifferent, we assume that it accepts the license.

If $z + q < \underline{z}(t)$ in the new-variety case, or if $q < 0$ in the creative-destruction case, entry does not occur and no license is agreed. We therefore focus on the states in which entry is feasible.

¹⁸If the agents can commit to licensing agreements, the results remain unchanged even if they play the enforcement game before the realization of imitation outcomes.

If the transfer leads to a new variety and the imitator enters, i.e., $z + q \geq \underline{z}(t)$, the innovator remains active regardless of the licensing decision. With a license, the innovator obtains $V(z, t) + \beta S(z + q, t)$ and the imitator obtains $W(t) + (1 - \beta) S(z + q, t)$. Without a license, the imitator uses the transferred knowledge without authorization and enters anyway; with probability β , enforcement excludes it from production and it receives $W(t)$. Hence its expected payoff is again $W(t) + (1 - \beta) S(z + q, t)$, so the offer is accepted.

If the transfer leads to creative destruction, the imitator can replace the incumbent only when $q \geq 0$. With a license, the incumbent is displaced and receives $W(t) + \beta S(z + q, t)$, while the imitator receives $W(t) + (1 - \beta) S(z + q, t)$. Without a license, the imitator again obtains $W(t) + (1 - \beta) S(z + q, t)$, whereas the innovator keeps its position with probability β and is displaced with probability $1 - \beta$. Its expected payoff is therefore $W(t) + \beta S(z, t)$. The innovator weakly prefers licensing whenever $S(z + q, t) \geq S(z, t)$, which holds for $q \geq 0$; the participation constraint is binding when $q = 0$ and strict when $q > 0$, so the license is accepted for all $q \geq 0$.

Aggregating over the imitation outcomes that lead to entry, the innovator's expected indirect net contribution from a match is

$$\begin{aligned} & \delta \int_{q \geq 0} [\beta S(z + q, t) - S(z, t)] dG(q) \\ & + (1 - \delta) \int_{z + q \geq \underline{z}(t)} \beta S(z + q, t) dG(q) \\ & = \beta S^e(z, t) - \tilde{\delta} S(z, t), \end{aligned}$$

where the first term captures creative-destruction matches and the second term captures new-variety matches. This is exactly the term that appears in equation (9). The imitator retains the complementary payoff

$$(1 - \beta) S^e(z, t),$$

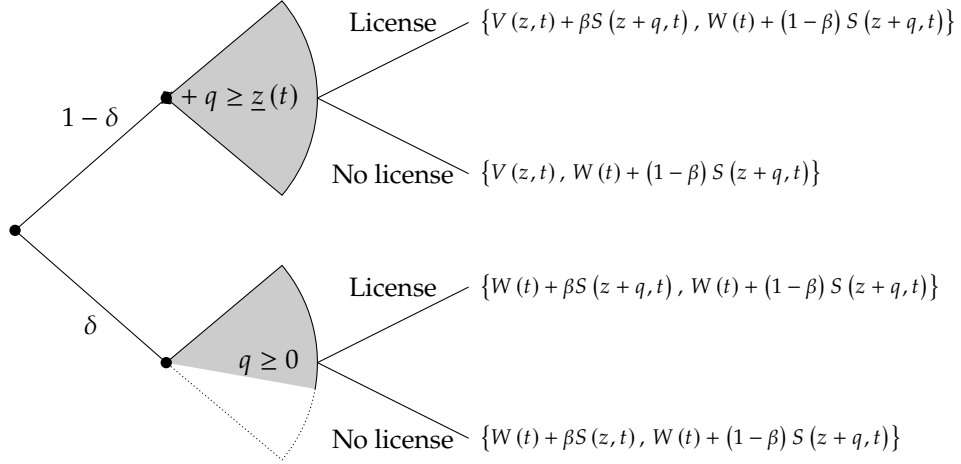
as in equation (10). If we express the innovator's payoff as a state-dependent share of the total surplus generated by a meeting, that total surplus can be written as

$$S^e(z, t) - \tilde{\delta} S(z, t).$$

Therefore, the corresponding implicit innovator bargaining weight is

$$\hat{\beta}(z, t) = \frac{\beta S^e(z, t) - \tilde{\delta} S(z, t)}{S^e(z, t) - \tilde{\delta} S(z, t)}.$$

Figure 4: Events and agent payoffs in the extended enforcement game



Under perfect learning, $q = 0$ almost surely. Therefore,

$$S^e(z, t) = S(z, t) \quad \text{and} \quad \tilde{\delta} = \delta.$$

Hence the implicit bargaining weight reduces to

$$\hat{\beta}(z, t) = \frac{\beta - \delta}{1 - \delta}.$$

A.3 Aggregation

Consider the problem of a representative final good producer that uses the intermediate goods as inputs. Let $P(t)$ denote the price of the final good and $p(z, t)$ the price of the intermediate good produced by a firm with productivity z . Given the aggregate production function in equation (5), the final good producer chooses the intermediate goods:

$$\max_{y(z, t)} P(t) Y(t) - \int p(z, t) y(z, t) dF(z, t).$$

The solution yields the demand for the intermediate goods:

$$\frac{y(z, t)}{Y(t)} = \left(\frac{p(z, t)}{P(t)} \right)^{-\eta}. \quad (25)$$

The zero-profit condition for the final good producer implies that the aggregate price is

$$P(t) = \left[\int p(z, t)^{1-\eta} dF(z, t) \right]^{\frac{1}{1-\eta}}. \quad (26)$$

Next, consider the profit maximization of a firm with productivity z . Given the innovation decision rule $\mu(z, t)$, the firm chooses price and labor input,

$$\pi(z, t) = \max_{p, \ell} p \left((1 - c(\mu(z, t))) e^z \right)^{\frac{1}{1-\eta}} \ell - w(t) \ell$$

subject to the demand function in equation (25). The firm charges a markup $\frac{\eta}{\eta-1}$ over the unit cost:

$$p(z, t) = \frac{\eta}{\eta-1} \left((1 - c(\mu(z, t))) e^z \right)^{\frac{1}{1-\eta}} w(t). \quad (27)$$

Substituting equation (27) into equation (26), we obtain that

$$P(t) = \frac{\eta}{\eta-1} \frac{w(t)}{Z(t)^{\frac{1}{\eta-1}}},$$

which is normalized to 1, implying that $w(t) = \frac{\eta-1}{\eta} Z(t)^{\frac{1}{\eta-1}}$. The firm's labor input is

$$\ell(z, t) = (1 - c(\mu(z, t))) e^z Z(t)^{-\frac{\eta}{\eta-1}} Y(t).$$

The labor market clearing condition, $\int \ell(z, t) dF(z, t) = L$, implies that

$$Y(t) = Z(t)^{\frac{1}{\eta-1}} L.$$

The firm's maximized profit from production is

$$\pi(z, t) = \Pi(t) (1 - c(\mu(z, t))) e^z, \text{ where } \Pi(t) = \frac{1}{\eta} Z(t)^{\frac{2-\eta}{\eta-1}} L.$$

A.4 Innovation Pattern

We formally establish the cross-sectional pattern of firm innovation in the following lemma.

Lemma 3 (Innovation Pattern). *The innovation intensity $\mu(z, t)$ increases in the level of pro-*

ductivity z and converges to an upper bound:

$$\lim_{z \rightarrow \infty} \mu(z, t) = \bar{\mu}(t).$$

Proof. Divide the entry surplus equation (8) by e^z :

$$\frac{S^e(z, t)}{e^z} = \delta \int_{q \geq 0} \left(\frac{V(z + q, t)}{e^z} - \frac{W(t)}{e^z} \right) dG(q) + (1 - \delta) \int_{z+q \geq \underline{z}(t)} \left(\frac{V(z + q, t)}{e^z} - \frac{W(t)}{e^z} \right) dG(q). \quad (28)$$

Similarly, divide the HJB equation (9) by e^z :

$$\begin{aligned} r(t) \frac{V(z, t)}{e^z} = \max_{\mu} & \left\{ \Pi(t) (1 - c(\mu)) + \mu \frac{V_z(z, t)}{e^z} + \frac{1}{2} \sigma^2 \frac{V_{zz}(z, t)}{e^z} + \frac{V_t(z, t)}{e^z} \right. \\ & \left. + \lambda(\alpha(t)) \left[\beta \frac{S^e(z, t)}{e^z} - \hat{\delta} \left(\frac{V(z, t)}{e^z} - \frac{W(t)}{e^z} \right) \right] \right\}. \end{aligned} \quad (29)$$

We guess and verify that the innovator's value function is asymptotically affine in e^z :

$$\lim_{z \rightarrow \infty} \frac{V(z, t)}{e^z} = v(t). \quad (30)$$

In equation (28), on the right-hand side, the component due to imitator's outside option relative to the innovator productivity $\frac{W(t)}{e^z}$ is decreasing in z . In particular, for firms on the knowledge frontier, this component converges to zero, $\lim_{z \rightarrow \infty} \frac{W(t)}{e^z} = 0$. Therefore,

$$\lim_{z \rightarrow \infty} \frac{S^e(z, t)}{e^z} = s^e v(t), \quad (31)$$

where $s^e = \delta \int_{q \geq 0} e^q dG(q) + (1 - \delta) \int e^q dG(q)$.

Differentiating equation (30) with respect to z and t , we obtain that

$$\lim_{z \rightarrow \infty} \frac{V_z(z, t)}{e^z} = \lim_{z \rightarrow \infty} \frac{V_{zz}(z, t)}{e^z} = v(t) \text{ and } \lim_{z \rightarrow \infty} \frac{V_t(z, t)}{e^z} = v_t(t). \quad (32)$$

Substituting the expressions in (31), (30) and (32) into equations (28) and (29), we obtain an ordinary differential equation that characterizes $v(t)$:

$$r(t) v(t) = \max_{\mu} \left\{ \Pi(t) (1 - c(\mu)) + \left(\mu + \frac{1}{2} \sigma^2 + \lambda(\alpha(t)) (\beta s^e - \hat{\delta}) \right) v(t) + v_t(t) \right\}.$$

The first-order condition with respect to μ at the limit is:

$$\Pi(t) c'(\bar{\mu}(t)) = v(t).$$

Thus we have verified that $V(z, t)$ is asymptotically affine in e^z .

We take the limit of the first-order condition (11):

$$\lim_{z \rightarrow \infty} \Pi(t) c'(\mu(z, t)) = v(t).$$

It must be that $\lim_{z \rightarrow \infty} \mu(z, t) = \bar{\mu}(t)$. □

A.5 Proof of Proposition 2

Endogenous innovation rate. Consider a BGP with a growth rate g . The interest rate is $r(t) = r = \rho + g$. The fraction of innovators $\alpha(t) = \alpha$. The surplus function satisfies $S(z + (\eta - 1)gt, t) = e^{gt}S(z)$. The value functions satisfy $V(z + (\eta - 1)gt, t) = e^{gt}V(z)$ and $W(t) = e^{gt}W$. The innovation threshold grows at rate $(\eta - 1)g$: $\underline{z}(t) = \underline{z} + (\eta - 1)gt$. The aggregate productivity grows at rate $(\eta - 1)g$: $Z(t) = e^{(\eta - 1)gt}Z$. The profit function satisfies $\Pi(t) = e^{(2 - \eta)gt}\pi$, where $\pi = \frac{1}{\eta}Z^{\frac{1}{\eta - 1} - 1}L$. Applying these properties, the entry surplus equation (8) and the HJB equations (9) and (10) become

$$S^e(z) = \delta \int_{q \geq 0} (V(z + q) - W) dG(q) + (1 - \delta) \int_{z + q \geq \underline{z}} (V(z + q) - W) dG(q) \quad (33)$$

$$\begin{aligned} \rho V(z) = \max_{\mu} & \left\{ \pi(1 - c(\mu))e^z + (\mu - (\eta - 1)g)V_z(z) + \frac{1}{2}\sigma^2 V_{zz}(z) \right. \\ & \left. + \lambda(\alpha) [\beta S^e(z) - \hat{\delta}(V(z) - W)] \right\} \end{aligned} \quad (34)$$

$$\rho W = \iota(\alpha)(1 - \beta)\mathbb{E}[S^e(z)].$$

The value matching and smooth pasting conditions (12) and (13) become

$$V(\underline{z}) = W \text{ and } V_z(\underline{z}) = 0.$$

To solve for the endogenous innovation decision, we apply the result in Lemma 3. According to equations (33) and (34), the functions $S^e(z)$ and $V(z)$ on the knowledge frontier is asymptotically affine in e^z :

$$\lim_{z \rightarrow \infty} \frac{S(z)}{e^z} = s^e v \text{ and } \lim_{z \rightarrow \infty} \frac{V(z)}{e^z} = v,$$

where

$$v = \pi \frac{1 - c(\bar{\mu})}{\rho + g - \bar{\mu} - \frac{1}{2}\sigma^2 - \lambda(\alpha)(\beta s^e - \hat{\delta})}.$$

Combining the expression for v above with the first-order condition (11) for innovation, we obtain equation (17) for characterizing the innovation upper bound $\bar{\mu}$.

Endogenous growth rate and distribution. Detrending the productivity distribution according to equation (15), we transform the KF equation (14) into an ordinary differential equation: $\forall z \geq \underline{z}$,

$$\begin{aligned} \lambda(\alpha) \left\{ \delta \int_0^\infty [f(z-q) - f(z)] dG(q) + (1-\delta) \int_{\underline{q}}^\infty f(z-q) dG(q) \right\} \\ + (\eta - 1) g f'(z) - \frac{\partial(\mu(z)f(z))}{\partial z} + \frac{1}{2}\sigma^2 f''(z) = 0. \end{aligned} \quad (35)$$

At the exit threshold $\underline{z}$, the density $f(\underline{z}) = 0$.

The stationary distribution does not allow for an analytical solution, due to the varying level of innovation intensity $\mu(z)$. However, using the result in Lemma 3 that the innovation intensity converges to a constant upper bound in the limit, we can solve analytically the asymptotic distribution in the tail. When $z \rightarrow \infty$, equation (35) converges to

$$\begin{aligned} \lambda(\alpha) \left\{ \delta \int_0^\infty [f(z-q) - f(z)] dG(q) + (1-\delta) \int_{\underline{q}}^\infty f(z-q) dG(q) \right\} \\ + ((\eta - 1) g - \bar{\mu}) f'(z) + \frac{1}{2}\sigma^2 f''(z) = 0. \end{aligned} \quad (36)$$

The asymptotic stationary distribution in the tail has a Pareto tail: for $z > \bar{z} \rightarrow \infty$, $f(z) \propto e^{-\zeta z}$, where the tail index ζ is characterized by

$$\frac{1}{2}\sigma^2 \zeta^2 + (\bar{\mu} - (\eta - 1) g) \zeta + \lambda(\alpha) \left[\delta \int_0^\infty (e^{\zeta q} - 1) dG(q) + (1-\delta) \int_{\underline{q}}^\infty e^{\zeta q} dG(q) \right] = 0.$$

We express the growth rate in terms of the tail index:

$$(\eta - 1) g = \bar{\mu} + \frac{1}{2}\sigma^2 \zeta + \frac{\lambda(\alpha)}{\zeta} \left[\delta \int_0^\infty (e^{\zeta q} - 1) dG(q) + (1-\delta) \int_{\underline{q}}^\infty e^{\zeta q} dG(q) \right].$$

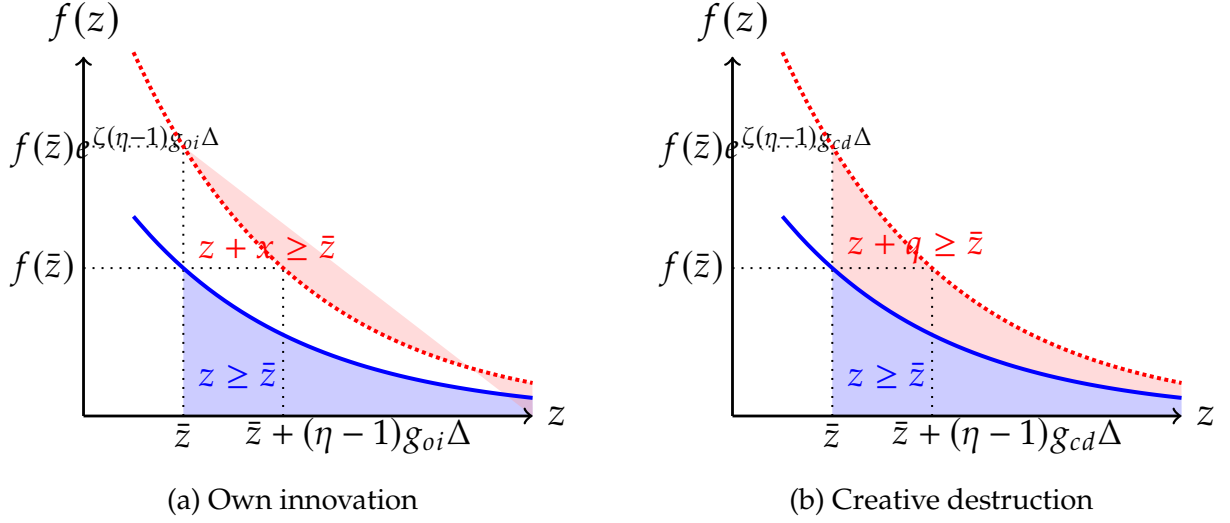


Figure 5: Events behind tail growth

Notes: Panels display how own innovation and creative destruction shift the tail of the productivity distribution.

There can potentially exist a continuum of equilibria associated with different growth rates and stationary distributions, as in [Luttmer \(2007\)](#) and [Benhabib et al. \(2021\)](#). When the initial distribution has a bounded support under Assumption 1, the economy converges to a unique BGP, which corresponds to the one with the lowest growth rate. This occurs when $\partial g / \partial \zeta = 0$, which implies equation (18).

A.6 Intuitive Growth Accounting

In this section we conduct two growth accounting exercises, which complement the BGP analysis above based on an intuitive accounting identity. We again highlight the endogenous objects that affect the growth outcome.

A.6.1 Growth in the Tail

The first growth accounting is according to the dynamics of the firms in the tail. We carry out the following thought experiment. Consider the detrended stationary distribution $F(z)$ on the BGP. We restrict our attention to the firms in the tail, i.e., those firms with productivity $z \geq \bar{z}$, where $\bar{z} \rightarrow \infty$. Their productivity aggregates to $\bar{Z} = \int_{\bar{z}}^{\infty} e^z dF(z)$. We then compute the change in aggregate productivity for the firms that remain or enter the tail region over a short time interval $\Delta \rightarrow 0$. We do so for the three types of innovation and learning events separately. The probability any of these events jointly happening is of higher orders of Δ , which without loss of generality can be disregarded.

First, we consider firms' own-innovation activities. One advantage of focusing on the firms in the tail is that their endogenous innovation effort approaches the constant limit, $\mu(z) \rightarrow \bar{\mu}$. Over the time interval Δ , the outcome of own innovation can thus be represented by a productivity increment drawn from a normal distribution: $x \sim N(\bar{\mu}\Delta, \sigma\sqrt{\Delta})$. For an innovator firm to remain in the tail or enter the tail, this increment must satisfy $z + x \geq \bar{z}$. These events lead to a shift in the productivity distribution depicted in panel (a) of Figure 5: accounting for the entry and exit into the tail while preserving the shape of the distribution, the distribution shifts forward by $(\eta - 1) g_{oi}\Delta$ amount, where g_{oi} denotes the growth due to own innovation. Naturally, the endogenous shape of the tail distribution ζ matters for how fast the mass in the tail grows. The eventual mass would be a factor $e^{\zeta(\eta-1)g_{oi}\Delta}$ of the starting mass. The growth in aggregate productivity can be computed according to

$$e^{\zeta(\eta-1)g_{oi}\Delta} = \frac{\int_{-\infty}^{\infty} \int_{\bar{z}-x}^{\infty} e^{z+x} dF(z) dH(x)}{\bar{Z}} = e^{\frac{1}{\eta-1}(\zeta\bar{\mu} + \frac{1}{2}\zeta^2\sigma^2)\Delta}. \quad (37)$$

The second equality in the equation above follows from the transformation for the normal distribution. We thus obtain that growth from own innovation is of rate $\bar{\mu} + \frac{1}{2}\zeta\sigma^2$:

$$g_{oi} = \frac{1}{\eta - 1} \left(\bar{\mu} + \frac{1}{2}\sigma^2\zeta \right). \quad (38)$$

Next, we consider the creative-destruction events. The probability an incumbent meets a learning firm who can potentially replace it is $\lambda(\alpha)\delta\Delta$. Similar to the proceeding discussion for own-innovation activities, we look at those learning firms that draw a quality $q \geq 0$ and $z + q \geq \bar{z}$ such that they land in the tail. The resulting change in the productivity distribution is described in panel (b) of Figure 5. Let g_{cd} denote the growth coming from these creative destruction events:

$$e^{\zeta(\eta-1)g_{cd}\Delta} - 1 = \frac{\lambda(\alpha)\delta\Delta \int_0^{\infty} \int_{\bar{z}-q}^{\infty} (e^{z+q} - e^z) dF(z) dG(q)}{\bar{Z}}. \quad (39)$$

Lastly, we consider the arrival of new varieties. For each incumbent variety, the probability of a new variety arriving is $\lambda(\alpha)(1 - \delta)\Delta$. To enter the right tail, the entrant draws a quality $q \geq \underline{q}$ such that $z + q \geq \bar{z}$. The implied shift in the productivity distribution mirrors the logic in Figure 5. The resulting output growth is

$$e^{\zeta(\eta-1)g_{nv}\Delta} - 1 = \frac{\lambda(\alpha)(1-\delta)\Delta \int_{\underline{q}}^{\infty} \int_{\bar{z}-q}^{\infty} e^{z+q} dF(z) dG(q)}{\bar{Z}}. \quad (40)$$

As we shrink the time interval to zero, the growth rate computed in equations (39) and (40) converge to the levels in the growth decomposition earlier. We summarize these equivalent measures in the following lemma.

Lemma 4. *Consider the shift in the distribution of firms in the right tail $\bar{z} \rightarrow \infty$ over a small time interval $\Delta \rightarrow 0$. The growth rates computed according to equations (37), (39), and (40) are characterized by the three components in the growth rate equation (16) in Proposition 2.*

Own innovation. For the numerator on the right-hand side of equation (37), we perform a change of variable: $z' = z + x$. The steady state density function satisfies $f(z' - x) = e^{\zeta x} f(z')$, for $z' - x \geq \bar{z}$. Thus:

$$\int_{-\infty}^{\infty} \int_{\bar{z}-x}^{\infty} e^{z+x} dF(z) dH(x) = \int_{-\infty}^{\infty} \int_{\bar{z}}^{\infty} e^{z'} dF(z' - x) dH(x) = \int_{\bar{z}}^{\infty} e^{z'} dF(z') \int_{-\infty}^{\infty} e^{\zeta x} dH(x).$$

Equation (37) simplifies to

$$e^{\zeta g_{oi}\Delta} = \left[\int_{-\infty}^{\infty} e^{\zeta x} dH(x) \right]^{\frac{1}{\eta-1}} = e^{\frac{1}{\eta-1}(\zeta\mu + \frac{1}{2}\zeta^2\sigma^2)\Delta},$$

from which we obtain the growth rate formula due to own innovation in equation (16).

Creative destruction. Finally, for the numerator on the right-hand side of equation (39), we perform a change of variable, $z' = z + q$. The steady state density function satisfies $f(z' - q) = e^{\zeta q} f(z')$, for $z' - q \geq \bar{z}$. Thus:

$$\int_0^{\infty} \int_{\bar{z}-q}^{\infty} e^{z+q} dF(z) d\tilde{G}(q) = \int_0^{\infty} \int_{\bar{z}}^{\infty} e^{z'} dF(z' - q) d\tilde{G}(q) = \int_{\bar{z}}^{\infty} e^{z'} dF(z') \int_0^{\infty} e^{\zeta q} d\tilde{G}(q).$$

It simplifies to

$$e^{\zeta g_{cd}\Delta} = \left[\lambda\delta(1 - G(0))\Delta \left(\int_0^{\infty} e^{\zeta q} \tilde{G}(q) - 1 \right) + 1 \right]^{\frac{1}{\eta-1}}.$$

Taking logs of the equation above, dividing both sides by Δ , and letting $\Delta \rightarrow 0$, we obtain the growth rate due to creative destruction in equation (16).

New varieties. Similarly, for the numerator on the right-hand side of equation (40), we perform a change of variable: $z' = z + q$. The steady state density function satisfies $f(z' - q) = e^{\zeta q} f(z')$, for $z' - q \geq \underline{z}$. Thus:

$$\int_{\underline{q}}^{\infty} \int_{\underline{z}-q}^{\infty} e^{z+q} dF(z) dG(q) = \int_{\underline{q}}^{\infty} \int_{\underline{z}}^{\infty} e^{z'} dF(z' - q) dG(q) = \int_{\underline{z}}^{\infty} e^{z'} dF(z') \int_{\underline{q}}^{\infty} e^{\zeta q} dG(q).$$

It simplifies to

$$e^{\zeta g_{nv}\Delta} = \left[\lambda(\alpha)(1-\delta)\Delta \int_{\underline{q}}^{\infty} e^{\zeta q} dG(q) + 1 \right]^{\frac{1}{\eta-1}}.$$

Taking logs of the equation above, dividing both sides by Δ , and letting $\Delta \rightarrow 0$, we obtain the growth rate due to new varieties in equation (16).

A.7 Proof of Lemma 1

We apply the Laplace transformation $\int_{\underline{z}}^{\infty} e^{-\xi z} x(z) dz$ to the KF equation (35) for the stationary distribution:

$$\left\{ \frac{1}{2} \sigma^2 \xi^2 + (\eta - 1) g \xi + \lambda(\alpha) \left[\delta \int_0^{\infty} (e^{-\xi q} - 1) dG(q) + (1 - \delta) \int_{\underline{q}}^{\infty} e^{-\xi q} dG(q) \right] \right\} \int_{\underline{z}}^{\infty} e^{-\xi z} f(z) dz - \xi \int_{\underline{z}}^{\infty} e^{-\xi z} \mu(z) f(z) dz = \frac{1}{2} \sigma^2 f'(\underline{z}) e^{-\xi \underline{z}}.$$

Letting $\xi = -1$:

$$\left\{ \frac{1}{2} \sigma^2 - (\eta - 1) g + \lambda(\alpha) \left[\delta \int_0^{\infty} (e^q - 1) dG(q) + (1 - \delta) \int_{\underline{q}}^{\infty} e^q dG(q) \right] \right\} \int_{\underline{z}}^{\infty} e^z f(z) dz + \int_{\underline{z}}^{\infty} e^z \mu(z) f(z) dz = \frac{1}{2} \sigma^2 f'(\underline{z}) e^{\underline{z}}. \quad (41)$$

Letting $\xi = 0$:

$$\lambda(\alpha)(1-\delta)\alpha = \frac{1}{2} \sigma^2 f'(\underline{z}). \quad (42)$$

Substituting equation (42) into equation (41), we obtain that the growth rate in the tail implied by KF equation.

A.8 Constant Innovation Benchmark

We now collect the closed-form expressions that obtain when the innovation intensity is exogenously fixed at $\mu(z) = \bar{\mu}$. In this case the Kolmogorov forward equation (35) reduces globally to the linear differential equation (36). The stationary density is exponential, $f(z) \propto e^{-\zeta z}$, with ζ characterized by equation (18). The aggregate growth rate simplifies to the BGP expression in (16), where the diffusion component becomes $\lambda(\alpha) D(\zeta)$ evaluated at the constant upper bound. These formulas provide the basis for the comparative statics highlighted in Section B and for the reference made in the main text.

A.9 Derivation of Example 1

Example 1 (Exponential quality distribution). Suppose the quality draw for knowledge diffusion follows an exponential distribution:

$$G(q) = 1 - e^{-\phi(q-\underline{q})}, \quad \forall q \geq \underline{q}.$$

Equation (16) for the growth rate becomes

$$D(\zeta) = \frac{1}{\zeta} \left[\delta e^{\phi \underline{q}} \frac{\zeta}{\phi - \zeta} + (1 - \delta) e^{\zeta \underline{q}} \frac{\phi}{\phi - \zeta} \right]. \quad (43)$$

Equation (18) that characterizes the Pareto tail index becomes

$$\lambda(\alpha) \left((1 - \delta) e^{\zeta \underline{q}} \frac{\phi (\phi - 2\zeta - \underline{q}\zeta (\phi - \zeta))}{\zeta^2 (\phi - \zeta)^2} - \delta e^{\phi \underline{q}} \frac{1}{(\phi - \zeta)^2} \right) = \frac{1}{2} \sigma^2. \quad (44)$$

Under the conditions in Lemma 3, when the quality draw follows an exponential distribution, we have $\int_{\underline{q}}^{\infty} e^q dG(q) = e^{\phi \underline{q}} \frac{\phi}{\phi - 1}$, $\int_0^{\infty} (e^q - 1) dG(q) = e^{\phi \underline{q}} \frac{1}{\phi - 1}$, $\int_{\underline{q}}^{\infty} e^{\zeta q} dG(q) = e^{\phi \underline{q}} \frac{\zeta}{\phi - \zeta}$, and $\int_{\underline{q}}^{\infty} (e^{\zeta q} - 1) dG(q) = e^{\phi \underline{q}} \frac{\phi}{\phi - \zeta}$. Given these expressions, we obtain that

$$D(1) = \delta e^{\phi \underline{q}} \frac{1}{\phi - 1} + (1 - \delta) e^{\phi \underline{q}} \frac{\phi}{\phi - 1}.$$

Further, $\int_0^{\infty} e^{\zeta q} q dG(q) = e^{\phi \underline{q}} \frac{\phi}{(\phi - \zeta)^2}$ and $\int_{\underline{q}}^{\infty} e^{\zeta q} q dG(q) = \frac{\phi}{\phi - \zeta} \left(\underline{q} + \frac{1}{\phi - \zeta} \right) e^{\zeta \underline{q}}$. Substituting these expressions into equation (18), we obtain equation (44).

All the heterogeneity is driven by imitation.

Lemma 5 (Deterministic Innovation). *In Example 1, suppose that the innovation outcome is deterministic, i.e., $\sigma = 0$. Specifically, the only source of randomness is imitation: $\underline{q} = 0$ and $\phi > 1$. Along the BGP, the economy grows at rate*

$$g = \frac{1}{\eta - 1} \left(\bar{\mu} + \frac{1}{\phi} \lambda(\alpha) \left(\sqrt{1 - \delta} + 1 \right)^2 \right), \quad (45)$$

and the endogenous stationary distribution has a Pareto tail index

$$\zeta = \phi \frac{\sqrt{1 - \delta}}{\sqrt{1 - \delta} + 1}. \quad (46)$$

Proof. When $\sigma = 0$, equation (44) for the tail index simplifies to

$$\frac{1 - \delta}{\zeta^2} - \frac{1}{(\phi - \zeta)^2} = 0.$$

Solving for the equation above, we obtain two roots: $\zeta_1 = \phi \sqrt{1 - \delta} / (\sqrt{1 - \delta} + 1)$ and $\zeta_2 = \phi \sqrt{1 - \delta} / (\sqrt{1 - \delta} - 1)$. Taking the positive root, we obtain the tail index in equation (46). Letting $\sigma = 0$ and substituting the tail index in (44) into (43), we obtain equation (45). $\square$

A.10 Proof of Lemma 2

When $\phi = \infty$ and $\underline{q} = 0$, equation (44) for the tail index simplifies to

$$\lambda(\alpha) \frac{1 - \delta}{\zeta^2} = \frac{1}{2} \sigma^2.$$

Solving for the equation above and taking the positive root, we obtain the tail index in equation (20). Letting $\phi = \infty$ and substituting the tail index in (20) into (43), we obtain equation (21).

A.11 Social Value Function

Let $V^s(z, t)$ denote the social value function associated with an innovating firm with productivity z at time t . Let $W^s(t)$ denote the social value function associated with a

copying firm at time t . The social value functions satisfy the following HJB equations:

$$\begin{aligned} r(t) V^s(z, t) = & \Pi(t) (1 - c(\mu)) e^z + \lambda(\alpha(t)) S(z, t) \\ & + \mu V_z^s(z, t) + \frac{1}{2} \sigma^2 V_{zz}^s(z, t) + V_t^s(z, t), \\ r(t) W^s(t) = & \phi(\alpha(t)) \int_{\underline{z}(t)} S(z, t) dF(z, t) + W_t^s(t). \end{aligned} \quad (47)$$

On the BGP, equation (47) simplifies to

$$\rho V^s(z) = \pi (1 - c(\mu)) e^z + \lambda(\alpha) S(z) + (\mu - (\eta - 1)g) V_z^s(z) + \frac{1}{2} \sigma^2 V_{zz}^s(z).$$

Taking the limit $z \rightarrow \infty$, we obtain $\lim_{z \rightarrow \infty} \frac{V^s(z)}{e^z} = \bar{V}^s$, where

$$\bar{V}^s = \frac{1 - c(\bar{\mu})}{\rho + (\eta - 1)g - \bar{\mu} - \frac{1}{2} \sigma^2 - \lambda(\alpha) D(1)}. \quad (48)$$

When q follows an exponential distribution, substituting $\int e^q dq = \frac{\phi}{\phi-1}$ into the expression above, we obtain the marginal social value function in equation (48).

B Social Planner's Problem

The problem in Section 5.3 is defined as follows. Given the initial distribution $F(z, 0)$, the planner chooses the path of policy $\{\beta(t)\}_{t \geq 0}$ and equilibrium outcome $\{S^e(z, t), W(t), \underline{z}(t), \alpha(t), \{V(z, t), \mu(z, t), F(z, t)\}_{z > \underline{z}(t)}\}_{t \geq 0}$ to maximize the welfare $\int_0^\infty e^{-\rho t} \log(C(t)) dt$ subject to the equilibrium conditions (8) to (14), final good market clearing condition $C(t) = Y(t)$, and the Euler equation. When we normalize the total expenditure $P(t)C(t)$ to 1, the aggregate profit is given by $\Pi(t) = 1/(\eta Z(t))$, and the Euler equation gives $r(t) = \rho$.

Let $u(t)$ denote the fraction of imitators at time t , where

$$u(t) = 1 - \alpha(t) \quad (49)$$

The KFE equation for innovators (14) and the definition of $\alpha(t)$ (7) implies the law of motion of the fraction of imitators:

$$u_t(t) = -m(\alpha(t), u(t)) + \frac{1}{2} \sigma^2 f_z(z, t). \quad (50)$$

We include (49) and (50) as a constraint of the planner instead of (7), because the interpretation of the costate variables will turn out to be more intuitive.

Now, we construct the current value Hamiltonian of this problem. To do this, we first define the sets of state variables $\mathbf{x}(t)$, control variables $\mathbf{y}(t)$, and costate variables $\boldsymbol{\psi}(t)$ as follows.

$$\begin{aligned}\mathbf{x}(t) &= \left[\{f(z, t), V(z, t)\}_{z > \underline{z}(t)} \right] \\ \mathbf{y}(t) &= \left[\beta(t), \underline{z}(t), \alpha(t), u(t), W(t), \{\mu(z, t)\}_{z > \underline{z}(t)} \right] \\ \boldsymbol{\psi}(t) &= \left[\psi^Z(t), \psi^\alpha(t), \psi^{bc}(t), \psi^u(t), \psi^w(t), \psi^{vm}(t), \psi^{vp}(t), \{\psi^f(z, t), \psi^v(z, t), \psi^\mu(z, t)\}_{z > \underline{z}(t)} \right]\end{aligned}$$

Then, the current value Hamiltonian is given by

$$\begin{aligned}\mathcal{H}(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\psi}(t)) &= \\ &\log(Z(t)) \\ &+ \psi^Z(t) \left\{ \int_{\underline{z}(t)}^{\infty} \left\{ (1 - c(\mu(z, t))) \frac{e^z}{Z(t)} f(z, t) \right\} dz - 1 \right\} \\ &+ \psi^\alpha(t) \{\alpha(t) + u(t) - 1\} \\ &+ \int_{\underline{z}(t)}^{\infty} \psi^f(z, t) \left[\lambda(\alpha(t)) \left\{ \begin{aligned} &\delta \int_0^{\infty} [f(z - q, t) - f(z, t)] dG(q) \\ &+ (1 - \delta) \int_{\underline{q}}^{\infty} f(z - q, t) dG(q) \end{aligned} \right\} \right. \\ &\quad \left. - \frac{\partial}{\partial z} (\mu(z, t) f(z, t)) + \frac{1}{2} \sigma^2 f_{zz}(z, t) \right] dz \\ &+ \psi^{bc}(t) f(\underline{z}(t), t) \\ &+ \psi^u(t) \left[-m(\alpha(t), u(t)) + \frac{1}{2} \sigma^2 f_z(z, t) \right] dt \\ &+ \int_{\underline{z}(t)}^{\infty} \psi^v(z, t) \left[\begin{aligned} &\rho V(z, t) - \frac{1}{\eta} (1 - c(\mu)) \frac{e^z}{Z(t)} - \mu V_z(z, t) - \frac{1}{2} \sigma^2 V_{zz}(z, t) \\ &- \lambda(\alpha(t)) \beta \left\{ \begin{aligned} &\delta \int_{q \geq 0} (V(z + q, t) - W(t)) dG(q) \\ &+ (1 - \delta) \int_{z+q \geq \underline{z}(t)} (V(z + q, t) - W(t)) dG(q) \\ &+ \lambda(\alpha(t)) \hat{\delta} (V(z, t) - W(t)) \end{aligned} \right\} \end{aligned} \right] dz \\ &+ \int_{\underline{z}(t)}^{\infty} \psi^w(t) [\rho W(t) - \iota(\alpha(t)) (1 - \beta) \mathbb{E}[S^e(z, t)]] dz \\ &+ \psi^{vm}(t) \{V(\underline{z}(t), t) - W(t)\} \\ &+ \psi^{vp}(t) V_z(\underline{z}(t), t) \\ &+ \int_{\underline{z}(t)}^{\infty} \psi^\mu(z, t) \left\{ \frac{1}{\eta} c'(\mu(z, t)) \frac{e^z}{Z(t)} - V_z(z, t) \right\} dz\end{aligned}$$

We rewrite the fourth term on the right-hand side of the Hamiltonian using integration by parts:

$$\begin{aligned}
& \int_{\underline{z}(t)}^{\infty} \psi^f(z, t) \left[\lambda(\alpha(t)) \left\{ \begin{aligned} & \delta \int_0^{\infty} [f(z-q, t) - f(z, t)] dG(q) \\ & + (1-\delta) \int_{\underline{q}}^{\infty} f(z-q, t) dG(q) \end{aligned} \right\} - \frac{\partial}{\partial z} (\mu(z, t) f(z, t)) + \frac{1}{2} \sigma^2 f_{zz}(z, t) \right] dz \\
&= \int_{\underline{z}(t)}^{\infty} \psi^f(z, t) \lambda(\alpha(t)) \left\{ \begin{aligned} & \delta \int_0^{\infty} [f(z-q, t) - f(z, t)] dG(q) \\ & + (1-\delta) \int_{\underline{q}}^{\infty} f(z-q, t) dG(q) \end{aligned} \right\} dz \\
&+ \int_{\underline{z}(t)}^{\infty} f(z, t) \left\{ \mu(z, t) \psi_z^f(z, t) + \frac{1}{2} \sigma^2 \psi_{zz}^f(z, t) \right\} dz \\
&- \psi^f(\infty, t) \mu(\infty, t) f(\infty, t) + \psi^f(\underline{z}(t), t) \mu(\underline{z}(t), t) f(\underline{z}(t), t) \\
&+ \frac{1}{2} \sigma^2 \left\{ -\psi_z^f(\infty, t) f(\infty, t) + \psi_z^f(\underline{z}(t), t) f(\underline{z}(t), t) + \psi^f(\infty, t) f_z(\infty, t) - \psi^f(\underline{z}(t), t) f_z(\underline{z}(t), t) \right\}
\end{aligned}$$

As a result of this procedure, it becomes straightforward to take the derivative with respect to $f(z, t)$. We also rewrite the sixth and last terms on the right-hand side of the Hamiltonian using integration by parts.

$$\begin{aligned}
& \int_{\underline{z}(t)}^{\infty} \psi^v(z, t) \left[\begin{aligned} & \rho V(z, t) - \frac{1}{\eta} (1-c(\mu)) \frac{e^z}{Z(t)} - \mu V_z(z, t) - \frac{1}{2} \sigma^2 V_{zz}(z, t) \\ & - \lambda(\alpha(t)) \beta \left\{ \begin{aligned} & \delta \int_{q \geq 0} (V(z+q, t) - W(t)) dG(q) \\ & + (1-\delta) \int_{z+q \geq \underline{z}(t)} (V(z+q, t) - W(t)) dG(q) \end{aligned} \right\} \\ & + \lambda(\alpha(t)) \hat{\delta} (V(z, t) - W(t)) \end{aligned} \right] dz \\
&= \int_{\underline{z}(t)}^{\infty} \psi^v(z, t) \left[\begin{aligned} & \rho V(z, t) - \frac{1}{\eta} (1-c(\mu)) \frac{e^z}{Z(t)} \\ & - \lambda(\alpha(t)) \beta \left\{ \begin{aligned} & \delta \int_{q \geq 0} (V(z+q, t) - W(t)) dG(q) \\ & + (1-\delta) \int_{z+q \geq \underline{z}(t)} (V(z+q, t) - W(t)) dG(q) \end{aligned} \right\} \\ & + \lambda(\alpha(t)) \hat{\delta} (V(z, t) - W(t)) \end{aligned} \right] dz \\
&+ \int_{\underline{z}(t)}^{\infty} V(z, t) \left\{ \frac{\partial}{\partial z} (\mu(z, t) \psi^v(z, t)) - \frac{1}{2} \sigma^2 \psi_{zz}^v(z, t) \right\} dz \\
&- \mu(\infty, t) \psi^v(\infty, t) V(\infty, t) + \mu(\underline{z}(t), t) \psi^v(\underline{z}(t), t) V(\underline{z}(t), t) \\
&- \frac{1}{2} \sigma^2 \left[-\psi_z^v(\infty, t) V(\infty, t) + \psi_z^v(\underline{z}(t), t) V(\underline{z}(t), t) + \psi^v(\infty, t) V_z(\infty, t) - \psi^v(\underline{z}(t), t) V_z(\underline{z}(t), t) \right]
\end{aligned}$$

and

$$\begin{aligned}
& \int_{\underline{z}(t)}^{\infty} \psi^{\mu}(z, t) \left\{ \frac{1}{\eta} c'(\mu(z, t)) \frac{e^z}{Z(t)} - V_z(z, t) \right\} dz \\
&= \int_{\underline{z}(t)}^{\infty} \left[\psi^{\mu}(z, t) \frac{1}{\eta} c'(\mu(z, t)) \frac{e^z}{Z(t)} + \psi_z^{\mu}(z, t) V(z, t) \right] dz \\
&+ \psi^{\mu}(\infty, t) V(\infty, t) - \psi^{\mu}(\underline{z}(t), t) V(\underline{z}(t), t)
\end{aligned}$$

As a result of this procedure, it becomes straightforward to take the derivative with respect to $f(z, t)$.

Given the Hamiltonian, the social planner solution satisfies the following optimality conditions

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{y}(t)} \mathcal{H}(\mathbf{x}(t), \mathbf{y}(t), \psi(t)) &= 0 \\
\frac{\partial}{\partial \mathbf{x}(t)} \mathcal{H}(\mathbf{x}(t), \mathbf{y}(t), \psi(t)) &= \rho \mathbf{x}(t) - \frac{d}{dt} \mathbf{x}(t)
\end{aligned}$$

and all the constraints. In the following, we focus on the stationary version of the social planner's problem.

Let us rename ψ^f as $\tilde{V}$ because the costate variable of the KFE for innovators corresponds to their social values. Similarly, let us rename ψ^u as $\tilde{W}$ because the costate variable of the law of motion for the fraction of imitators corresponds to the social value of an imitator. Given the social values $\tilde{V}$ and $\tilde{W}$, let us define the social match surplus:

$$\tilde{S}(z) = \delta \int_0^{\infty} [\tilde{V}(z+q) - \tilde{V}(z)] dG(q) + (1-\delta) \int_{\max\{q, \underline{z}-z\}}^{\infty} [\tilde{V}(z+q) - \tilde{W}] dG(q). \quad (51)$$

Rearranging the optimality conditions, we obtain the HJB equations for the social value functions of innovators and imitators:

$$\begin{aligned}
\rho \tilde{V}(z) &= \frac{1}{\eta-1} Z^{\frac{2-\eta}{\eta-1}} L(1-c(\mu)) e^z + (\mu(z) - (\eta-1)g) \tilde{V}_z(z) + \frac{1}{2} \sigma^2 \tilde{V}_{zz}(z) \\
&+ \lambda(\alpha) \left\{ \tilde{S}(z) - (1-\omega) \mathbb{E}[\tilde{S}(z)] + \frac{\psi^w}{1-\alpha} ((1-\beta)S(z) - (1-\omega) \mathbb{E}[S(z)]) \right. \\
&\left. + (1-\omega) \delta \int_{\underline{z}}^{\infty} \frac{\psi^v(\tilde{z})}{\alpha} (V(\tilde{z}) - W) d\tilde{z} \right\}, \quad (52)
\end{aligned}$$

$$\rho \tilde{W} = \gamma(\alpha) \left\{ (1-\omega) \mathbb{E}[\tilde{S}(z)] + \frac{\psi^w}{1-\alpha} (\beta-\omega) \mathbb{E}[S(z)] - (1-\omega) \delta \int_{\underline{z}}^{\infty} \frac{\psi^v(\tilde{z})}{\alpha} (V(\tilde{z}) - W) d\tilde{z} \right\}. \quad (53)$$

with value matching and smooth pasting conditions:

$$\tilde{V}(\underline{z}) = \tilde{W}, \quad (54)$$

$$\tilde{V}_z(\underline{z}) = 0. \quad (55)$$

These HJB equations, value matching conditions, and smooth pasting conditions are analogous to the private ones but include the external effects on matching. These matching externalities include the value of the match for both parties and the crowding out externalities due to matching.

In addition to the value functions and all the constraints for the planner, the remaining four conditions obtained from the optimality conditions are required to pin down the multipliers $\psi^v(z)$, $\psi^\mu(z)$, ψ^w , and the optimal bargaining power β :

$$\psi^\mu(z) = \frac{\tilde{V}_z(z) - \frac{\eta}{\eta-1} V_z(z)}{\Pi c''(\mu(z)) e^z} f(z). \quad (56)$$

$$\begin{aligned} 0 = \lambda(\alpha) & \left\{ \beta \left[\delta \int_0^{z-\underline{z}} \psi^v(z-q) dG(q) + (1-\delta) \int_{\underline{q}}^{z-\underline{z}} \psi^v(z-q) dG(q) \right] - \tilde{\delta} \psi^v(z) \right\} \\ & + \lambda(\alpha) (1-\beta) \frac{\psi^w}{1-\alpha} \left[\delta \int_0^{z-\underline{z}} f(z-q) dG(q) + (1-\delta) \int_{\underline{q}}^{z-\underline{z}} f(z-q) dG(q) \right] \\ & + (\eta-1) g \psi_z^v(z) - \frac{\partial(\mu(z) \psi^v(z))}{\partial z} + \frac{1}{2} \sigma^2 \psi_{zz}^v(z) - \psi_z^\mu(z). \end{aligned} \quad (57)$$

$$\int_{\underline{z}}^{\infty} \psi^v(z) dz + \psi^w = 0. \quad (58)$$

$$\int_{\underline{z}}^{\infty} S^e(z) \frac{\psi^v(z)}{\alpha} dz = \frac{\psi^w}{1-\alpha} \mathbb{E}[S^e(z)]. \quad (59)$$

In summary, the social planner allocation is characterized by the equilibrium conditions (8) to (14) and the optimality conditions (51) to (59)

C Scalable-Volatility Innovation Process

For the case with scalable-volatility, most of the equations characterizing the equilibrium are unchanged. We modify a few equations where the volatility specification matters.

The HJB equation (9) for the innovating firm's value function becomes

$$r(t) V(z, t) = \max_{\mu} \left\{ \Pi(t) (1 - c(\mu)) e^z + \beta \lambda(\alpha(t)) S(z, t) \right. \\ \left. + \mu V_z(z, t) + \frac{1}{2} \sigma(\mu)^2 V_{zz}(z, t) + V_t(z, t) \right\}$$

The optimality condition (11) for innovation now takes into account the effects on not only the upward drift but also the volatility:

$$\Pi(t) c'(\mu(z, t)) e^z = V_z(z, t) + \frac{1}{2} \sigma^2 V_{zz}(z, t).$$

The KF equation (14) for the distribution of firm productivity is modified to

$$f_t(z, t) = \lambda(\alpha(t)) \left\{ \delta \int_0^\infty [f(z - q, t) - f(z, t)] dG(q) + (1 - \delta) \int_{\underline{q}}^\infty f(z - q, t) dG(q) \right\} \\ - \frac{\partial(\mu(z, t) f(z, t))}{\partial z} + \frac{1}{2} \frac{\partial^2(\sigma^2 \mu(z, t) f(z, t))}{\partial z^2}. \quad (60)$$

Along the BGP, the growth rate due to own innovation (38) derived from the KF equation (60) is modified to

$$g_{oi} = \frac{1}{\eta - 1} \left(\bar{\mu} + \frac{1}{2} \sigma^2 \bar{\mu} \zeta \right)$$

where the upper bound of innovation $\bar{\mu}$ is characterized by

$$c'(\bar{\mu}) = \frac{(1 + \frac{1}{2} \sigma^2) (1 - c(\bar{\mu}))}{\rho + (\eta - 1) g - \bar{\mu} - \frac{1}{2} \sigma^2 \bar{\mu} - \beta D(1)}.$$

Finally, the endogenous stationary distribution, $\forall z > \underline{z}$, has a Pareto tail with the shape parameter

$$\zeta = \frac{\sqrt{2(1 - \delta) \lambda(\alpha)}}{\sigma \sqrt{\bar{\mu}}}.$$